My Notes

3 chapters, 57 sections - compiled 2026-10-10

1. Geology 3

- Colluvium and Alluvium 4
- Epigenesis 5
- Groups, Formations and members 6
- Hypogene-Supergene 8
- MISC 9
- Relative Age 11
- Strike and dip 13

2. Geostatistics 16

- Accuracy-Precision 17
- Block Kriging and Uncertainty For data spacing 20
- Block size (Clayton) 21
- Categorical Variogram and Transition Probability 23
- Change of Support Equations 25
- Covariance of Residuals 26
- Cross and Direct Variogram 27
- Data Spacing 29
- Dual Kriging 31
- Ergodic vs Non-ergodic 32
- Estimation (Random Variable Concept) 33
- Estimation and Multivariate Relationships 34
- Factors affecting Uncertainty 35
- Gibbs Sampling 36
- Heteroscedasticity and Heterogeneity 37
- Importance of Nugget in Estimation 39
- Interpretation of Cross Validation Plots 40
- Kriging Compare to Other estimation techniques 41
- LCS vs E-type 42
- PDF, Mean, Variance, covariance 43
- Point Scale to SMU Scale 44
- Proportional Effect and Data Spacing 45
- RMSP Trend removal steps 47

Search Anisotropy and Variogram Search 48
Spatial Bootstrap and Conditioning Data 50
Spike and NS transform 51
Super Secondary 53
Trend Intuition 55
Trend Removal 57
Variogram Non-Stationary 59

3. Mining 60

Angle Conventions (Leapfrog Support) 61
Assay Types 62
Compositing 63
Cutoff grade and curves 65
Density Calculations 66
Dilution and Volume Variance Correction 67
Drilling 68
Engineering Terms and Units 69
Estimation and Conditional Bias 70
Estimation Models 72
Gaussian Distribution and Scaling 73
Kriging Plan 75
Metallurgy 77
Mineral Scale 78
Monthly, Quarterly and Annual upscaling 79
Net Smelter Return 81
Open Pit Mining 82
Regular spacing to 3 hole spacing 83
Resource Classification 84
Scale, combined grade and Variability 85

Chapter 1

Geology

7 sections

Colluvium and Alluvium

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes

Colluvium is a general term for loose, unconsolidated sediments deposited at the base of hill slopes by rainwash, sheetwash, slow continuous downslope creep, or a combination of these processes.

Alluvium accumulates in and next to river channels, whereas colluvium is deposited by mass-wasting processes acting on hill slopes.

Additional notes

	Colluvium	Alluvium
Transport agent	Gravity, sheetwash, creep (short distance)	Flowing water in streams and rivers
Sorting	Poor; mixed grain sizes	Moderate to good; sorted by flow energy
Clast shape	Angular to sub-angular	Rounded to sub-rounded
Layering	Massive or crudely layered	Stratified, often cross-bedded
Typical setting	Slope toes, talus aprons	Floodplains, channels, terraces, alluvial fans

- **Eluvium (residuum):** weathered material that has stayed in place, on top of its parent rock (no transport). Laterites and saprolite are residual.
- **Talus (scree):** coarse, angular colluvium that accumulates by rockfall below cliffs.
- **Alluvial fan:** a cone-shaped alluvial deposit where a steep stream enters a flatter valley.
- Together with glacial (till), aeolian (loess) and marine deposits, these are forms of **transported regolith** (cover).

Practical relevance in mining and exploration: (1) **Placer deposits** of gold, tin and diamonds form in alluvium, and also in eluvium and colluvium near the source. (2) Transported cover masks bedrock geochemistry, so soil sampling over colluvium or alluvium can give false anomalies or miss buried targets; distinguishing residual from transported regolith is key. (3) Cover is usually modelled as a separate, often waste or low-grade, domain with its own density.

Epigenesis

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes

In geology, epigenesis is the modification of minerals in rocks after they form. Epigenesis often happens near the Earth's surface due to exposure to oxygen, carbon dioxide and water.

Clarification: "epigenesis" is used in two related senses. (1) In sedimentary petrology it means changes after diagenesis, typically late, near-surface alteration such as weathering-related oxidation and cementation. (2) In ore geology, **epigenetic** describes mineralization introduced into a host rock *after* the host rock formed, which can happen at great depth, not only near the surface.

Additional notes

	Syngenetic	Epigenetic
Timing	Ore forms at the same time as the host rock	Ore introduced after the host rock formed
Geometry	Usually stratabound / stratiform, conformable with layering	Often cross-cutting (veins, stockworks, breccias, replacements)
Examples	VMS, SEDEX, banded iron formation, placers, evaporites	Orogenic gold veins, porphyry Cu, epithermal Au-Ag, MVT Pb-Zn, skarns, Carlin-type Au

- **Diagenesis** covers the physical and chemical changes in sediments during and after burial (compaction, cementation, recrystallization) at low temperature and pressure, before metamorphism.
- The boundary between diagenetic and epigenetic is not sharp, and terminology varies between authors.
- Evidence for epigenetic timing: veins cutting bedding or foliation, alteration halos around fluid conduits, and ore minerals replacing earlier minerals.

Practical relevance: whether mineralization is syngenetic or epigenetic affects how it is modelled. Syngenetic ores tend to follow stratigraphy (so model within stratigraphic surfaces), while epigenetic ores follow structures and fluid pathways (so model veins, faults and alteration envelopes, with structurally controlled anisotropy).

Groups, Formations and members

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes

Hierarchy: Group → Formations → Members.

Formation: the fundamental stratigraphic unit, a body of rock with a consistent set of physical characteristics. In other words, a formation is a series of beds distinct from the beds above and below it.

Group: two or more related formations can be classified together as a group. A group may be up to several thousand metres thick and may represent deposition over a few to tens of millions of years.

> Clarification: thickness and duration vary widely between groups. The figures above are typical, not part of the definition. A group is defined by the shared characteristics of its formations.

Member: a formation may be divided into members, each with a specific and distinct lithology.

Additional notes

These are **lithostratigraphic** units: they are defined by rock type (lithology) and the boundaries are physical contacts, which can be time-transgressive. They differ from **chronostratigraphic** units (eonothem, erathem, system, series, stage), which are defined by time.

Rank	Description
Supergroup	Two or more related groups (and/or formations).
Group (subgroup)	Two or more formations with significant shared features.
Formation	The primary formal unit; must be mappable at the surface or traceable in the subsurface at common map scales.
Member	A lithologically distinct part of a formation; not necessarily mappable everywhere.
Bed / flow	The smallest formal unit: a single distinctive layer (bed) in sedimentary rocks, or a single flow in volcanic rocks.

- **Naming:** formal names combine a geographic name with a rank or lithology, e.g. "Morrison Formation" or "Navajo Sandstone".
- **Lens / tongue:** a lens is a lens-shaped body inside a formation, and a tongue is part of a unit that projects into a neighbouring one. These are common at facies transitions.
- Units are defined at a **type section** (stratotype), where the unit and its boundaries are formally described.

Practical relevance: drill hole logs and geological models usually code formations and members as categorical domains. Consistent use of the hierarchy helps when defining estimation domains and contact (boundary) analysis.

Hypogene/Supergene

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes

In ore deposit geology, hypogene processes occur deep below the Earth's surface and tend to form deposits of primary minerals. Supergene processes occur at or near the surface and tend to form secondary minerals.

Hypogene rocks include deep-forming rocks (plutonic and metamorphic) and also, but rarely, volcanic rocks.

> Clarification: in modern usage, "hypogene" mostly describes *processes and ore minerals* formed by ascending (usually hydrothermal or magmatic) fluids, not rock types as such. The rock-type usage is older and less common.

Additional notes

Supergene enrichment happens when descending meteoric water (rain/groundwater) oxidizes sulphides above the water table, dissolves metals (especially Cu), and re-precipitates them lower down. It is most important for porphyry copper deposits and some Ag, Au, Ni and Zn deposits.

Zone (top to bottom)	Characteristics	Typical minerals
Leached cap / gossan	Metals removed; iron oxides left behind	Limonite, goethite, hematite, jarosite
Oxide zone	Oxidized ore above the water table	Malachite, azurite, chrysocolla, cuprite
Supergene enrichment blanket	Near the paleo water table; secondary sulphides replace primary ones; grades can be several times hypogene grade	Chalcocite, covellite, digenite
Hypogene (primary) zone	Unaltered primary mineralization	Chalcopyrite, bornite, pyrite, molybdenite

- Enrichment needs: pyrite to generate acid, a slowly falling water table, permeable host rock, and enough time (often tectonic uplift in arid to semi-arid climates).
- The gossan colour and minerals are used in exploration to infer what sulphides lie below.

> Practical relevance: supergene, oxide and hypogene zones usually have very different grades, densities, metallurgical recoveries (oxide ores often go to heap leach, sulphides to flotation) and geotechnical properties. They are normally modelled as separate estimation domains, with careful treatment of the contacts between them.

MISC

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes (corrected)

Conformable: parallel strata with a similar geological history, deposited in sequence without interruption (no significant gap in time). The opposite is an *unconformity*.

Chert: a hard, fine-grained sedimentary rock made of microcrystalline (or cryptocrystalline) quartz, SiO_2.

> Correction: chert is not ordinary quartz grains. It is made of microscopic quartz crystals (and sometimes chalcedony or opal). It forms by accumulation of siliceous organisms (radiolaria, diatoms, sponge spicules) or by chemical replacement, e.g. flint nodules in chalk. Varieties include flint, jasper (red, iron-rich) and novaculite. Cherts can host or be associated with gold, for example in banded iron formation and jasperoid settings, but chert is not generally gold-bearing.

Pyrite: also known as fool's gold, the most abundant sulphide mineral; iron disulphide, FeS_2. It is brassy yellow and cubic, with a greenish-black streak.

> Additional: pyrite often carries "invisible" gold in its crystal structure (e.g. Carlin-type and orogenic deposits). Oxidation of pyrite produces sulphuric acid, which drives acid rock drainage and supergene leaching.

Schist: a medium- to coarse-grained metamorphic rock with pronounced schistosity: platy minerals (micas, chlorite, talc) are large enough to see easily with a hand lens and are aligned, so the rock splits into thin flakes or plates (foliated).

> Additional: schist forms by regional metamorphism, typically at medium grade. The increasing-grade sequence for mudrocks is slate → phyllite → schist → gneiss.

Dolerite: a fine- to medium-grained, dark grey to black mafic intrusive igneous rock.

> Additional: dolerite (called diabase in North America) has the same composition as basalt and gabbro (plagioclase plus pyroxene). It crystallizes at shallow depth (hypabyssal), typically in dykes and sills, and often shows an ophitic texture.

Hornfels: the group name for contact metamorphic rocks that have been baked and hardened by the heat of intrusive igneous masses. They are massive, hard, splintery and in some cases exceedingly tough and durable.

> Additional: hornfels is fine-grained and non-foliated, because it forms without directed stress. It occurs in the contact aureole around intrusions, and skarns can form in carbonate rocks in similar settings.

Anticline / syncline

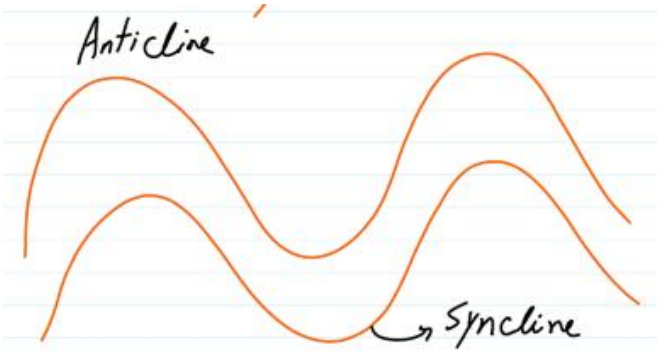


Figure 1. Anticline (upward fold) and syncline (downward fold).

- **Anticline:** a fold that is convex upward, with the **oldest** rocks in its core; the limbs dip away from the axis.
- **Syncline:** a fold that is concave upward, with the **youngest** rocks in its core; the limbs dip toward the axis.

Correction / precision: strictly, "anticline" and "syncline" refer to the age of the rocks in the core. If the stratigraphic order is unknown (e.g. in metamorphic terrains), use the purely geometric terms *antiform* and *synform*. Other fold terms: hinge, axial plane, limbs, plunge, and monocline (a single step-like bend).

Practical relevance: anticlines are classic structural traps for oil and gas, and fold hinges often localize mineralization (e.g. saddle reefs in orogenic gold).

Relative Age

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes

1) Original horizontality: beds of sediment deposited in water form as horizontal or nearly horizontal layers.

2) Superposition: within a sequence of undisturbed sedimentary rocks, the layers get younger from bottom to top.

3) Cross-cutting relationships: a disrupted pattern is older than the cause of the disruption.

> Clarification: in other words, any feature that cuts across rocks (a fault, dyke, vein or erosion surface) is younger than the rocks it cuts.

Additional notes

Relative dating orders events (older/younger) without assigning ages in years. Absolute (numerical) ages come from radiometric methods such as U-Pb, Ar-Ar or Rb-Sr. Most of these principles were set out by Nicolas Steno (1669) and James Hutton (late 1700s).

- **Lateral continuity:** layers originally extend sideways until they thin out or hit the edge of the basin. Matching layers on either side of a valley or fault were once continuous.
- **Inclusions (included fragments):** fragments of one rock inside another are older than the host. For example, xenoliths in a granite are older than the granite, and pebbles in a conglomerate are older than the conglomerate.
- **Faunal succession:** fossil assemblages succeed one another in a definite order, so they can be used to correlate rocks of the same age between areas.
- **Baked contacts:** an intrusion bakes (contact-metamorphoses) the rock it intrudes, so the baked rock is older.

Unconformities

An unconformity is a buried erosion or non-deposition surface that represents missing time (a gap in the record).

Type	Description
Angular unconformity	Tilted or folded older strata overlain by younger flat-lying strata.
Disconformity	Parallel strata above and below, but with an irregular erosion surface between them.
Paraconformity	Parallel strata with no obvious erosion surface; the gap is recognized from fossils or dating.

Nonconformity	Sedimentary rocks deposited on eroded igneous or metamorphic basement.

Practical relevance: in resource modelling, these principles drive the interpretation of geological sequences, the order in which domains and contacts are modelled (e.g. later dykes cutting mineralization), and whether a post-mineral intrusion should be modelled as barren.

Strike and dip

OneNote notebook RMS / section Geology. Original notes (lightly corrected) followed by additional reference notes.

Original notes (refined)

Geologists use their own convention to describe the orientation of planes (bedding, faults, veins, mineralized zones).

Strike: the azimuth (compass direction, measured clockwise from North) of a horizontal line on the plane. It is the line where the plane intersects a horizontal surface.

> Refinement: the original note described strike as "the azimuth of the bedding on the plan view". That is only exact for the trace of the plane on a horizontal surface; on sloping topography the outcrop trace is not the strike. Strike is always defined by a horizontal line.

Dip: the angle between the plane and the horizontal, measured perpendicular to the strike in the direction of steepest descent. Dip is always between 0 and 90°.

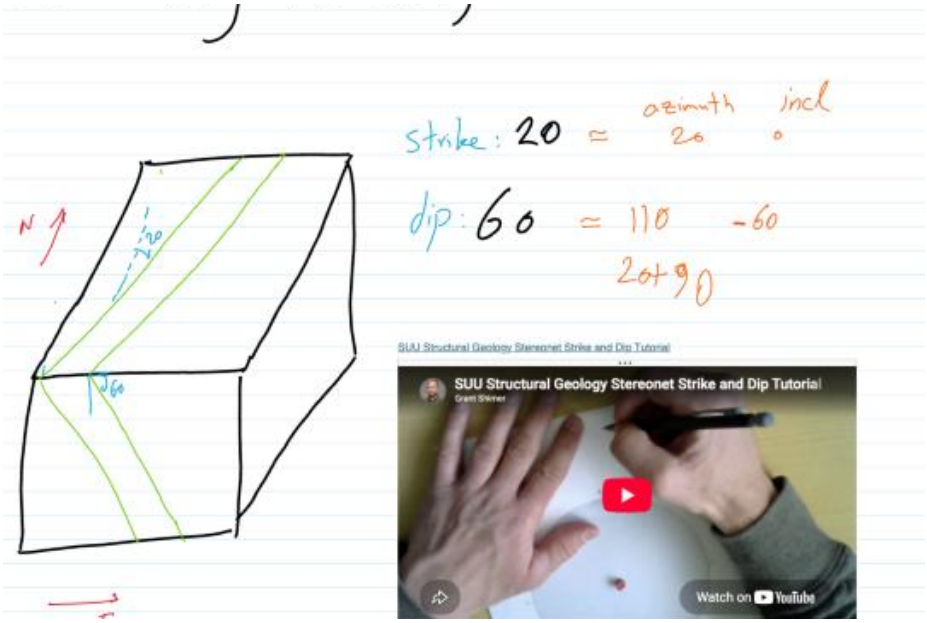


Figure 1. Example block: strike 20 and dip 60, i.e. dip direction 110 (= 20 + 90) and inclination −60.

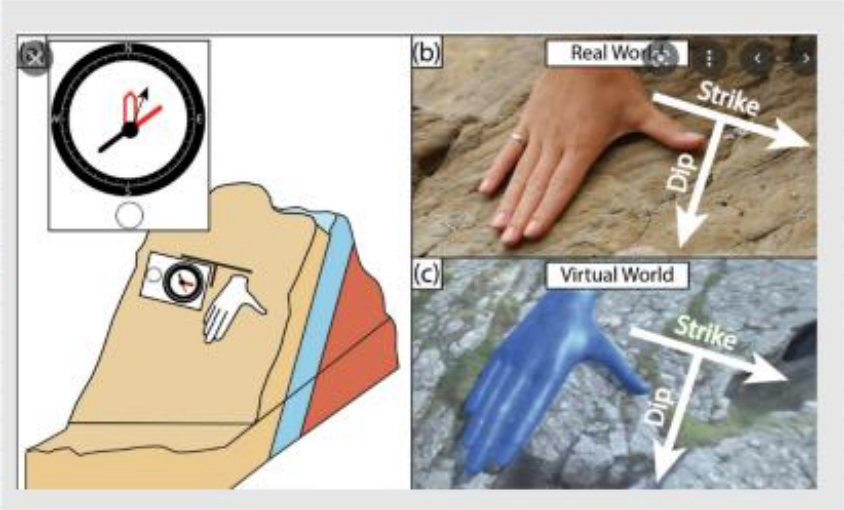


Figure 2. Strike and dip on an outcrop (pasted reference figure).

Normal = dip × strike (cross product) → normal vector (pole) of the mineralization plane

Precision: the order of the cross product sets the sign. With unit vectors along the dip (pointing down-dip) and along the strike (right-hand rule), dip × strike gives the upward-pointing normal, and strike × dip gives the downward one. Either way, the normal is perpendicular to the plane.

Strike and dip often describe the mineralization plane. The major direction of continuity within the plane can also be given as azimuth and dip:

azimuth = strike + 90 (right-hand rule), inclination = −dip

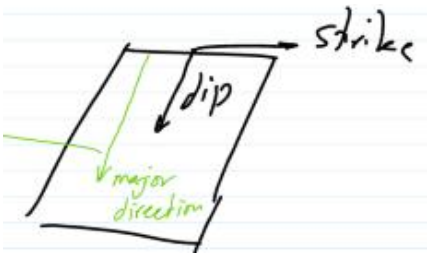


Figure 3. Strike, dip and the major direction along the plane.

Additional notes

- **Right-hand rule (RHR):** record the strike so that the plane dips to the *right* when you face along the strike. Then dip direction = strike + 90°. Without this rule, the dip quadrant must be written out, e.g. 020/60 SE.
- **Dip direction / dip** notation, e.g. **110/60**, is often less ambiguous and is preferred in many software packages and structural logging sheets. Strike 020, dip 60 (RHR) = dip direction 110, dip 60.
- **Quadrant notation**, e.g. N20E 60SE, is common in older literature.

- **Apparent dip:** in a section that is not perpendicular to strike, the dip looks shallower: $\tan(\text{apparent dip}) = \tan(\text{true dip}) \times \sin(\beta)$, where β is the angle between the strike and the section line.
- **Lines** (lineations, fold axes, drill holes) use **trend and plunge** instead: the azimuth of the line's horizontal projection, and its angle below horizontal.
- **Pole to plane:** the normal plotted on a stereonet. Pole trend = dip direction − 180 (+360 if needed), and pole plunge = 90 − dip.

Practical relevance: variogram and search ellipsoids in geostatistical software are often defined by azimuth/dip/rake (or pitch). Converting field strike/dip correctly, and checking the convention of each package (see the Angle Conventions note), avoids misoriented models.

Chapter 2

Geostatistics

30 sections

Accuracy vs Precision

OneNote notebook RMS / section Geostat - Thursday, April 29, 2021, 7:28 AM. Transcribed from handwritten notes.

In a set of measurements, **accuracy** is closeness to a specific value, while **precision** is the closeness of the measurements to each other.

One approach is to observe and assess accuracy and precision separately, as shown in this figure:

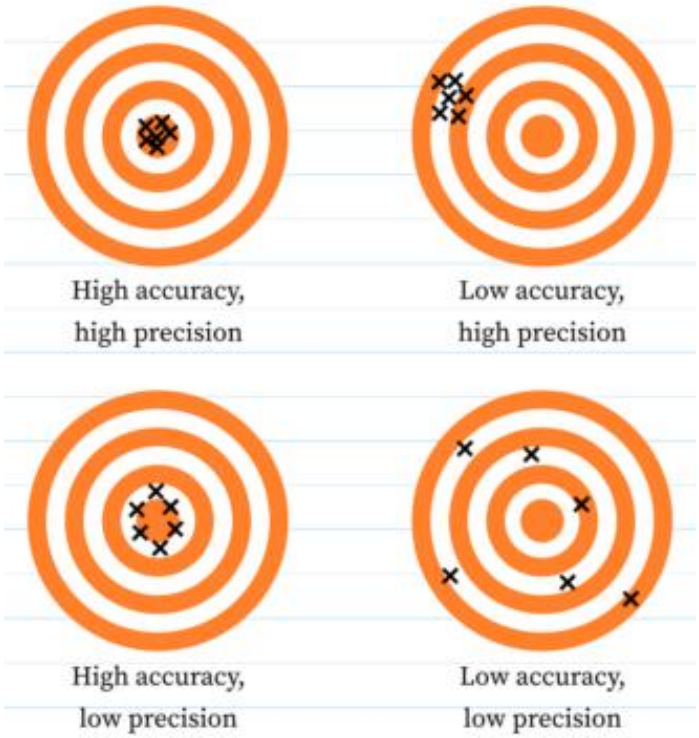


Figure 1. Accuracy and precision assessed separately.

Alternatively, ISO describes accuracy as a combination of both. So, high accuracy needs both high trueness and precision.

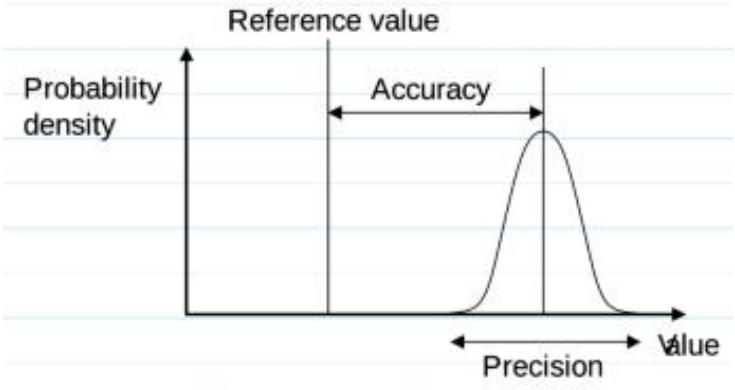


Figure 2. Accuracy (offset from the reference value) and precision (spread of the distribution).

> Accuracy is the closeness of measurement results to the true value; precision is the degree of closeness of repeated measurements under unchanged conditions.

- Accuracy → trueness → statistical bias
- Precision → repeatability → systematic error

Although these two words can be synonyms, they are contrasted in engineering and science. In statistics, the terms bias and variability are preferred instead of accuracy and precision.

- Low accuracy → high bias → low trueness
- Low precision → high variability → low repeatability

Geostat Context

In the context of geostatistics, we are interested in checking the distribution of uncertainty, which is concerned with the assessment of conditional probabilities.

- Local distributions or CCDF models may be derived from L realizations, directly from indicator kriging, or defined by a Gaussian mean, variance and transformation.

In order to check the "goodness" of a probabilistic model, specific definitions of accuracy and precision are proposed that are not closely/intuitively related to the concepts described above.

The proposed definitions are based on the actual fraction of true values falling within symmetric probability intervals of varying width *p*:

– A probability distribution is **accurate** if the fraction of true values falling in the *p* interval exceeds *p* for all *p* in [0, 1].

– The **precision** of an accurate probability distribution is measured by the closeness of the fraction of true values to *p* for all *p*.

Consider the following model of uncertainty that provides a local ccdf:

> $F^*(u_i; z(u_i) \mid n(u_i))$ → distribution of the random variable Z at location u_i, conditioned to $n(u_i)$ nearby data.

> If the true value is at the median, F would be 0.5.

For each probability interval *p*, a symmetric probability interval PI can be defined:

> $p\text{-PI} = (p_{low}, p_{upp}) = ((1 - p)/2 , (1 + p)/2)$
>
> e.g. 0.9-PI = (0.05, 0.95) → covers 90% of states.
>
> $\xi(u_i; p) = 1$ if $F(u_i; z(u_i) \mid n(u_i)) \in (p_{low}, p_{upp})$; 0 otherwise

$\xi_{avg}(p) = E[\xi] = (1/n)\ \Sigma_{i=1..n}\ \xi(u_i; p)$ → proportion of locations where the true value falls within the symmetric p-PI (written as ξ with an overbar in the original notes)

p	(p_{low}, p_{upp})	ξ	E[ξ]
10	(0.45, 0.55)	[0 1 1 ... 0]	0.12
⋮	⋮	⋮	⋮
90	(0.05, 0.95)	[1 1 1 0 ... 1]	0.92

Based on what was discussed above, the ccdfs or distributions of uncertainty are **accurate** if $\xi_{avg}(p) \geq p\ \forall p$. The **precision** is defined as the closeness of $\xi_{avg}(p)$ to p when we have accuracy (i.e. $\xi_{avg}(p) \geq p$).

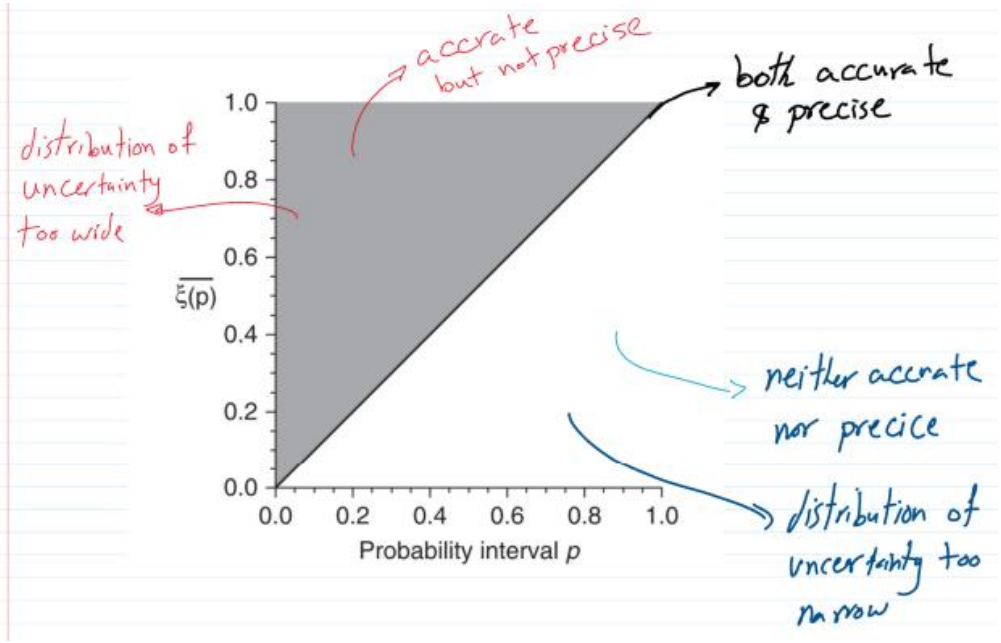


Figure 3. Accuracy plot of $\xi_{avg}(p)$ vs probability interval p. Shaded area: accurate but not precise (distribution of uncertainty too wide). On the 1:1 line: both accurate and precise. Below the line: neither accurate nor precise (distribution of uncertainty too narrow).

- For a **wide** distribution, one way is to increase continuity (e.g. variogram ranges) to increase conditioning and decrease uncertainty.
- In contrast, for a **narrow** distribution of uncertainty, decreasing the range of spatial correlation will increase the width of the local distributions of uncertainty.

If two different probabilistic models are equally accurate and precise, we should use the one with lower uncertainty. The average conditional variance at all locations can be calculated to identify the model with the highest local uncertainty:

$$\text{"U"} = (1/N)\ \Sigma_{i=1..N}\ \sigma^2(u_i)$$

Block Kriging and Uncertainty For data spacing

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Block kriging is often used to get estimates at a target block/support/SMU. Block kriging is based on C(V, u) (V: block, u: point) → the average of the covariance between the sample and all the points inside the target volume.

$$Z^*(u, V) = \Sigma_{\alpha=1..n} \lambda_\alpha Z(u_\alpha, V)$$

It can be shown that the block kriging estimate is the same as estimating at the discretized locations and taking the average. However, the kriging variance is of particular interest to assess uncertainty at block scale.

With OK in original units, the kriging variance has no mathematical meaning, but it is directly related to uncertainty and changes with conditioning. So, we can consider a nominal panel/block and use block ordinary kriging with reasonable discretization to get the kriging variance that represents the local uncertainty at that panel/block scale.

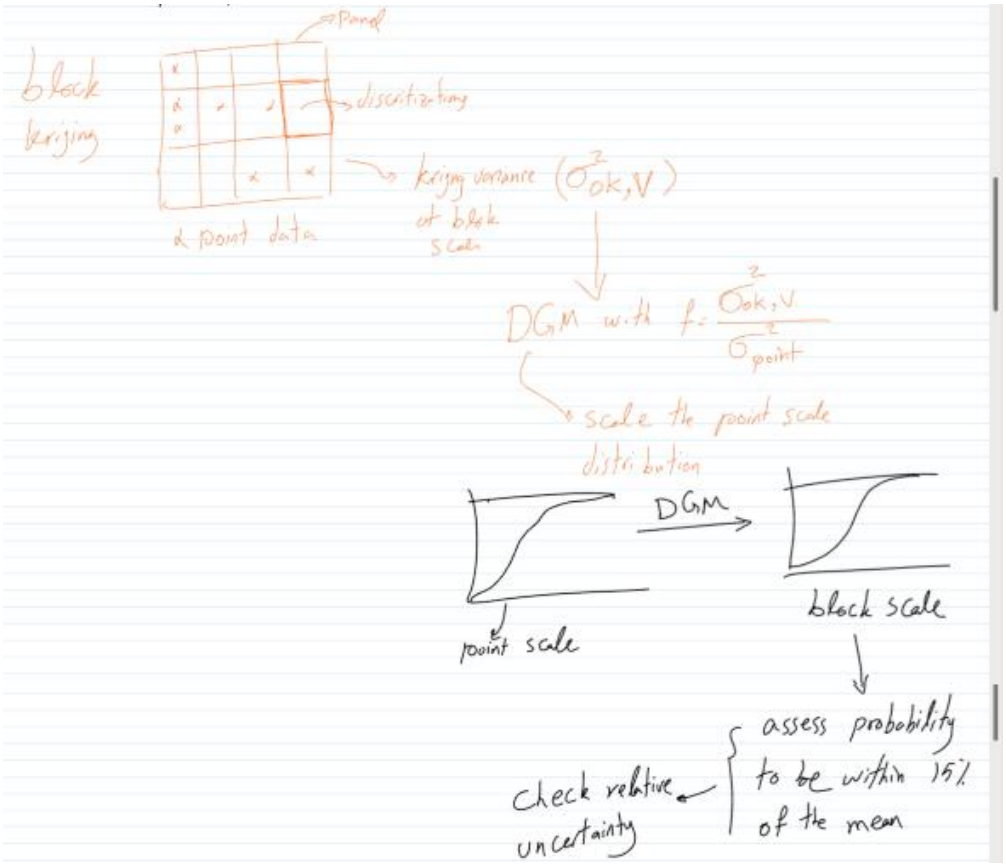


Figure 1. Block kriging workflow: block kriging with point data and a discretized panel gives the kriging variance at block scale ($\sigma^2_{OK,V}$); a DGM with $f = \sigma^2_{OK,V} / \sigma^2_{point}$ scales the point scale distribution to the block scale, which is used to assess the probability of being within 15% of the mean (relative uncertainty).

Block size (Clayton)

It is my pleasure to address this support question. I get to pontificate on one of my favorite subjects and get paid for it; life does not get much better than that! Let me give some pointed comments, then become more general:

1. A block size less than 1/3 to 1/4 of the data spacing does not resolve any new information. This is backed up by the earliest work of David (see below). The work of Yaroslav Vasylchuk and myself (lots of CCG references and papers) would confirm that 1/4 is a good ratio. In fact, Michel David also locked in on exactly 1/4 – one quarter of data spacing.

2. It is critical to note that since those days (look at M. David's text below) until now, most knowledgeable geostatisticians understand that block estimates for block sizes smaller than one quarter the data spacing are not wrong; they just provide the same estimate and provide no additional information. Small block estimates are not wrong or bad, they just do not resolve additional spatial features.

3. Larger blocks, however, smear out spatial information and do not provide precision in geological boundaries. So, given that vanishingly small block estimates are the same as larger block estimates, it is mind boggling that people are concerned about small block estimates. Who believes that small block estimates actually represent small scale variability!?! Must be some naïve newcomers to geostatistics that attribute super powers to kriging?

4. High resolution simulated realizations averaged to different block sizes are a different story – they would show different variability. Perhaps some folks believe kriging can provide the same information as simulation? The short answer to this is "no".

5. The SMU size is based on mining equipment, mining method, mining operations, direction of mining, data to support grade control, visual distinction of ore and waste, blast movement and its predictability, supervision,… The work of Neufeld, Leuangthong and myself on this subject is a good reference.

6. The SMU size absolutely does not depend on the exploration data spacing or the declustering diagnostic plot! That is just crazy. Think about it – mine selectivity at the time of mining depends on whether we drilled closer or more widely during exploration? Really?! We are going to make the best decisions at the time of mining regardless of what we knew earlier.

7. Any DGM calculations should be aimed at the target SMU size – that is correct.

8. Calibrating and tuning an estimated model to the anticipated DGM results for a rational SMU size makes sense. Once again – nothing to do with the exploration drill spacing! Sheesh…

9. Now, I am becoming more general.

10. Small block estimates (less than one quarter the data spacing) are exactly the same as larger parent block estimates, so they cannot be terrible – they are the same! They do not represent the variability at that scale and they may trick some naïve model users to believe the block estimates are correct – a misguided confidence in a block model.

11. So, if small block estimates are exactly the same as larger block estimates (provided all block sizes are less than one quarter the data spacing), what is the problem? There is none. Just don't trust estimates that are too small. They are not wrong – our misguided belief that they may be right is wrong. A large block estimate is also wrong.

12. Theoretically (or, at least, analytically), larger blocks are always better. The kriging variance, kriging efficiency, slope of regression and all other metrics of performance increase when the block size increases. Once again, if the estimates are the same – how can small blocks be worse!?

13. The present-day data spacing should have no influence of the appropriate SMU size (determined by parameters described above); however, the present data spacing can influence the block model, that is, the block model size need not be at the SMU size. The variability implicit to the block model should represent the target SMU, but that can be achieved with different block sizes.

14. Future data spacing at the time of mining could have a very significant influence on the representative SMU size. Not saying that the SMU size should be one third of future data spacing because there are many other factors (as mentioned), but the final model used for actual mine operations at the time of mining should be based on a high-resolution model (one quarter the data spacing).

15. A simulation study could be used to calibrate the SMU size (see the study in CCG by Neufeld, Leuangthong and myself). Usually, the scale of selectivity is chosen subjectively based on many factors.

Categorical Variogram and Transition Probability

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Indicator variogram

An indicator is a concept of assigning a binary state to a random variable. The expected value or estimated indicator corresponds to the probability of the state.

$I(u; k) = 1$ if $Z(u) = k$; 0 otherwise → for categorical variables

$I(u; Z = z_c) = 1$ if $Z(u) \leq z_c$; 0 otherwise → for thresholds on continuous variables

Note: the expected value of an indicator is the probability of the binary event that it represents.

The indicator variogram is defined as:

$$\gamma_{ij}(h) = 1/(2N(h)) \Sigma \{ [I_i(u) - I_i(u+h)] [I_j(u) - I_j(u+h)] \}, i, j \in [1, \ldots, K]$$

The indicator cross variograms ($i \neq j$) are often ignored, as they show extreme continuity at short lags and cannot be modeled by conventional co-regionalization models.

$$\gamma_k(h) = 1/(2N) \Sigma (I(u; k) - I(u+h; k))^2, (i = j = k), k \in [1, \ldots, K]$$

As can be seen above, the indicator variogram of one category measures its transition probability from existing to not existing over a Euclidean distance (h).

Kriging

The most common type of kriging used for indicators is SK. The SK estimates for indicators are often used to represent the conditional probability of a category; however, there is no theoretical support for that.

$$I_k^{SK}(u_0) - P_k = \Sigma_{\alpha=1..n} \lambda_{\alpha,k} [I_k(u_\alpha) - P_k] \forall k \in [1, \ldots, K]$$

$$I_k^{LVM1}(u_0) - P_k(u_0) = \Sigma_{\alpha=1..n} \lambda_{\alpha,k} [I_k(u_\alpha) - P_k(u_\alpha)] \forall k \in [1, \ldots, K]$$

$$I_k^{LVM2}(u_0) - P_k(u_0) = \Sigma_{\alpha=1..n} \lambda_{\alpha,k} [I_k(u_\alpha) - P_k(u_\alpha)] \forall k \in [1, \ldots, K]$$

The kriging equations above are often used to quantify the conditional probability of each category. This is followed by order relation corrections (sum to 1 and non-negative) before forming a cumulative probability mass function that can be used for Monte Carlo simulation.

Mean and variance

$$E\{I_i\} = P_i \ \forall i \in [1, \ldots, K]$$

$Var[I_i] = E[I_i^2] - E[I_i]^2 = P - P^2 = P(1 - P)$ → often used as the sill for an indicator variogram

Transition probability

Closely related to the indicator variogram:

$t_{ij}(h) = P(I_j(u+h) = 1 \mid I_i(u) = 1) = P(I_j(u+h) = 1, I_i(u) = 1) / P(I_i(u) = 1), i, j \in [1, \ldots, K]$

$T_{ij}(h) = [t_{ij}(h)] \leftarrow$ transition probability matrix of rank K

Transition probabilities are often calculated along drill holes, as the data density is higher and the shortest lag separations are well informed. The transition matrices are often asymmetric ($t_{ij}(h) \neq t_{ji}(h)$) and are calculated upwards and downwards before being averaged to populate the matrix:

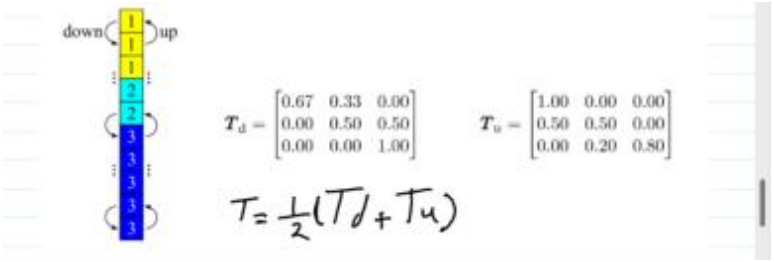


Figure 1. Downward and upward transition matrices along a drill hole; $T = ½ (T_d + T_u)$.

If the bivariate distribution of I(u), I(u+h) depends only on the separation vector (equation 2.30) and the categorical proportion is stationary (equation 2.31), i.e. first- and second-order stationarity, the indicator variogram can be rewritten in terms of transition probabilities (equation 2.32):

$$\mathbf{1}_i(\boldsymbol{u}+\boldsymbol{h}) = 1 \text{ and } \mathbf{1}_j(\boldsymbol{u}) = 1\}, \quad i,j \in \{1,\ldots,B\} \text{ and } \forall \boldsymbol{u} \in \mathcal{A} \qquad (2.30)$$
$$\mathbf{1}_i(\boldsymbol{u})\,\mathbf{1}_j(\boldsymbol{u}+\boldsymbol{h})\}$$

$$= \mathrm{P}\{\mathbf{1}_i(\boldsymbol{u}) = 1\} = \mathrm{E}\{\mathbf{1}_i(\boldsymbol{u})\}, \quad i \in \{1,\ldots,B\} \text{ and } \forall \boldsymbol{u} \in \mathcal{A} \qquad (2.31)$$

$$_j(\boldsymbol{h}) = p_i\left(t_{i,j}(0) - \frac{1}{2}[t_{i,j}(\boldsymbol{h}) + t_{i,j}(-\boldsymbol{h})]\right), \quad i,j \in \{1,\ldots,B\} \qquad (2.32)$$

$\gamma_{ii}(h) = P_i (1 - t_{ii}(h))$

The variogram is related to the overall transition of one category, rather than the conditional one. In that sense, $t_{ij}(h)$ is equivalent to the covariance, since its higher value indicates a higher probability of having j at (u+h) conditional to i at (u).

Direct variogram

$\gamma_{ii}(h) = ½ E\{ [I(u) - I(u+h)]^2 \}$

$= ½ [E\{I^2(u)\} + E\{I^2(u+h)\} - 2E\{I(u) \cdot I(u+h)\}]$

$= ½ [2P_i - 2E\{I_i(u) \cdot I_i(u+h)\}]$

$= P_i [1 - E\{I_i(u) \cdot I_i(u+h)\} / P_i] = P_i (1 - t_{ii}(h))$

Change of Support Equations

OneNote notebook RMS / section Geostat (page title spelled "Equetions" in OneNote). Transcribed from handwritten notes.

$$\sigma^2(\cdot, A) = \sigma^2(V, A) + \sigma^2(\cdot, V)$$

- $\sigma^2(\cdot, A)$: total point scale variance
- $\sigma^2(V, A)$: variance of blocks of size V within the domain
- $\sigma^2(\cdot, V)$: point scale variance within a block

$$f = \sigma^2(V, A) / \sigma^2(\cdot, A) = (\sigma^2(\cdot, A) - \sigma^2(\cdot, V)) / \sigma^2(\cdot, A) = 1 - \sigma^2(\cdot, V) / \sigma^2(\cdot, A)$$

The page includes a pasted textbook worked example: 12 point values grouped into 2 × 3 blocks. In that example the total point variance splits into the variance of the block averages plus the average variance of points within their blocks:

$$40 = 2.5 + 37.5$$

> Reference definition (paraphrased): dispersion variance is an average squared difference that explicitly states the support of the individual values and the support of the mean they are compared to.

It can be shown that:

$$\sigma^2(\cdot, A) = 1/(2n^2) \Sigma_{i=1..n} \Sigma_{j=1..n} (v_i - v_j)^2 = 1/n^2 \Sigma_i \Sigma_j (v_i - v_j)^2/2 = 1/n^2 \Sigma_i \Sigma_j \gamma(h_{ij})$$

(the average representation of $(v_i - v_j)^2/2$ is $\gamma(h_{ij})$).

Therefore:

$$\sigma^2(\cdot, V) = 1/n^2 \Sigma_{i=1..n} \Sigma_{j=1..n} \gamma(h_{ij}) = \bar{\gamma}(V), \text{ for } (i, j) \text{ with } h_{ij} \in V$$



Figure 1. Discretize the volume.

$$f = 1 - \bar{\gamma}(V) / \sigma^2 \quad (\bar{\gamma}(V): \text{average point scale variogram within the block})$$

Alternatively

$$Z_V = (1/n) \Sigma_{i \in V} Z(x_i) \text{ (variable at block scale)}$$

$$Var(Z_V) = 1/n^2 \Sigma_{i=1..n} \Sigma_{j=1..n} C_{ij}$$

The variance of a linear combination is the double sum of the covariances.

$$f = (1/n^2 \Sigma_i \Sigma_j C(h_{ij})) / \sigma^2 = C(V) / \sigma^2$$

Covariance of Residuals

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Note that the covariance of X, Y is equal to the covariance of $(X - m_x)$, $(Y - m_y)$, i.e. the residuals:

$$\mathrm{Cov}\{X, Y\} = E[(X - m_x)(Y - m_y)] = E[X \cdot Y] - m_x \cdot m_y$$

$$X' = X - m_x \rightarrow E[X'] = E[X] - m_x = 0$$

$$Y' = Y - m_y \rightarrow E[Y'] = E[Y] - m_y = 0$$

$$\mathrm{Cov}\{X', Y'\} = E[(X' - m_{x'})(Y' - m_{y'})] = E[X' \cdot Y'] - m_{x'} \cdot m_{y'} = E[X' \cdot Y']$$

$$= E[(X - m_x)(Y - m_y)] = E[X \cdot Y] - m_x \cdot m_y$$

Thus, the covariance does not change after translation, but it changes with scaling.

> Note: the variogram, like the mean, changes with any linear transformation, including translation and scaling.

Final note: with residuals, the covariance can be calculated as $E\{X' \cdot Y'\}$, since the means are known to be zero ($\rightarrow$ stationarity). But for the original variables, the mean is required to calculate the covariance.

Cross and Direct Variogram

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Direct

$$\gamma_{ZZ}(h) = 1/(2N(h)) \Sigma_{i=1..N(h)} [Z(u_i) - Z(u_i+h)]^2$$

For the direct variogram, this is the moment of inertia about the 45° line:

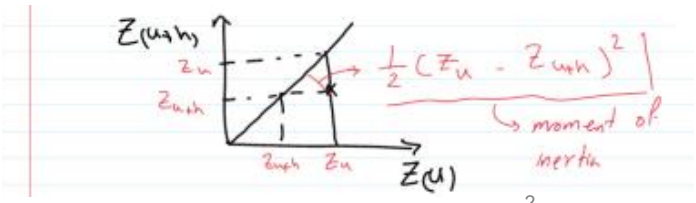


Figure 1. h-scatterplot of Z(u+h) vs Z(u): each pair contributes ½ $(Z_u - Z_{u+h})^2$, the moment of inertia about the 45° line.

$$C_{ZZ}(h) = 1/N(h) \Sigma Z(u) \cdot Z(u+h) - m_{Z(u)} \cdot m_{Z(u+h)}$$

$$\Rightarrow C_{ZZ}(h) = C_{ZZ}(0) - \gamma_{ZZ}(h) \text{ (first- and second-order stationarity)}$$

As can be seen from the above equations:

$$\gamma_{ZZ}(h) = \gamma_{ZZ}(-h)$$

$$C_{ZZ}(h) = C_{ZZ}(-h) \leftarrow \text{auto-covariance, or covariance function}$$

$$\rho_{ZZ}(h) = C_{ZZ}(h) / (\sigma_{Z(u)} \sigma_{Z(u+h)})$$

The page also has a pasted reference excerpt on the linear model of coregionalization (LMC): each random function $Z_i(u)$ is modeled as a linear combination of standard independent random factors $Y^l_k(u)$ that share L + 1 covariance structures, i.e. $Z_i(u) = \Sigma_{l=0..L} \Sigma_{k=1..n_l} a^l_{i,k} Y^l_k(u) + \mu_i$, and $C_{i,j}(h) = \Sigma_{l=0..L} \Sigma_k a^l_{i,k} a^l_{j,k} c_l(h)$.

Cross (two different variables)

For the cross variogram, the 45° line and the moment of inertia definitions are not relevant anymore. Instead, the variogram is defined to be consistent with the covariance. The concept of direct is extended to cross:

$$C_{ZY}(h) = 1/N(h) \Sigma Z(u) \cdot Y(u+h) - m_{Z(u)} \cdot m_{Y(u+h)}$$

$$\gamma_{ZY}(h) = 1/(2N(h)) \Sigma (Z(u) - Z(u+h)) \cdot (Y(u) - Y(u+h)) \rightarrow \gamma_{ZY} = C_{AB}$$

As can be seen above, for the cross variogram and covariance:

$$C_{ZY}(h) \neq C_{ZY}(-h) \text{ and } C_{ZY}(h) \neq C_{YZ}(h) \text{ (known as the lag effect)}$$

$$\gamma_{ZY}(h) = \gamma_{ZY}(-h)$$

$$\text{but } C_{ZY}(h) = C_{YZ}(-h)$$

If we expand the cross variogram:

$$\gamma_{ZY}(h) = ½ [E\{Z(u)Y(u)\} - E\{Z(u)Y(u+h)\} + E\{Z(u+h)Y(u+h)\} - E\{Y(u)\cdot Z(u+h)\}]$$

$$= C_{ZY}(0) - (C_{ZY}(h) + C_{YZ}(h)) / 2$$

In most implementations, the lag effect is ignored, as it is not significant in most cases in practice, and therefore:

$\gamma_{ZY}(h) = C_{ZY}(0) - C_{ZY}(h)$, assuming that $C_{ZY}(h) \cong C_{YZ}(h)$

Data Spacing

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

1) Variogram continuity and uncertainty at different scales

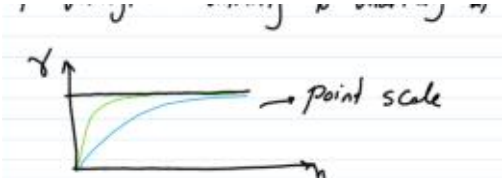


Figure 1. Point scale variograms with different continuity.

Higher continuity results in higher local accuracy, and at the same time simulated values are more similar inside a production panel.

Is the conclusion that more continuity results in higher uncertainty at larger scales also influenced by data spacing? **Yes, it is.**

→ Depending on the conditioning effect, higher continuity can result in lower or higher local uncertainty.

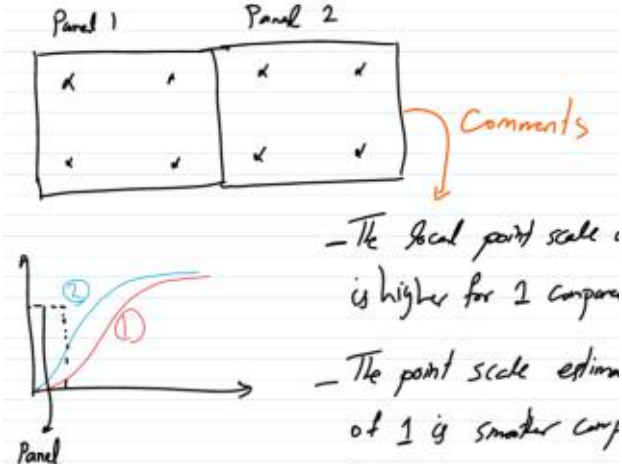
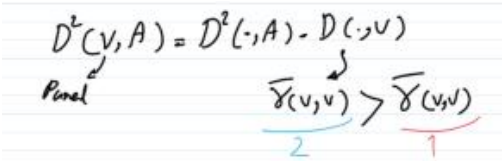


Figure 2. Two panels with the same data configuration, and two variograms: (1) less continuous, (2) more continuous.

Comments:

- The local point scale accuracy is higher for 1 compared to 2.
- The point scale estimation of 1 is smoother compared to 2.
- The variability of panels is expected to be lower for case 2 compared to 1, based on the volume variance:

  $D^2(v, A) = D^2(\cdot, A) - D^2(\cdot, v)$, with $\bar{\gamma}(v, v)_2 > \bar{\gamma}(v, v)_1$



However, the local uncertainty of panels that is a result of our limited data is quantified by multiple realizations of the same panel, rather than variability across panels. This local uncertainty is the target of data spacing studies. The panel scale uncertainty is influenced by a combination of factors, including variogram continuity and conditioning.

Dual Kriging

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Dual kriging is a matrix operation trick that enables us to calculate the weights once in a "global kriging" setup where all data points are used.

$$\Sigma_{\beta=1..n} b_\beta C_{\alpha\beta} = z_\alpha - m_\alpha \,,\ \alpha = 1, \ldots, n$$

→ equation system to get the weights, based on $C_{\alpha\beta}$ and the residuals.

$$Z^*(x_0) - m = \Sigma_{\alpha=1..n} \lambda_\alpha [Z(x_\alpha) - m] = \Sigma_{\alpha=1..n} b_\alpha C(x_\alpha - x_0)$$

→ estimator based on a weighted sum of covariances.

Note: with a global kriging setup and using all conditioning data, the weights are calculated once (b_β), and in order to get the estimator value, only $C(x_\alpha - x_0)$ needs to be found for each unsampled location.

Ergodic vs Non-ergodic

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Ergodic

In mathematics, ergodicity means that the system/model in question will eventually visit all its possible states.

In the context of geostatistics, this means a realization samples the space of uncertainty close to uniformly, visiting all extreme cases observed from the reference distribution.

In general, as variogram ranges increase, with a fixed data number and configuration, the global uncertainty increases due to increased redundancy.

If the variogram range is bigger than half of the domain size, the realizations start to show behaviour known as non-ergodic fluctuations, where realizations do not sample the space of uncertainty uniformly, which can result in inflated global uncertainty. This is because the realizations' distributions can be more different, as they cannot visit all highs and lows uniformly.

> Note: ergodic fluctuations are expected to decrease as the number of conditioning data increases.

Estimation (Random Variable Concept)

OneNote notebook RMS / section Geostat. The page contains a pasted textbook excerpt; this is a summary of it, not a copy.

The excerpt contrasts estimation in **reality** with the **conceptual (random function) model** used to justify it:

	Reality	Conceptual model
Data	A sample data set of n values $v_1, \ldots, v_n$	A stationary random function of n + 1 random variables $V(x_0)$, $V(x_1)$, …, $V(x_n)$
Estimate	A specific value $\hat{v}_0 = \Sigma\, w_i \cdot v_i$, a weighted linear combination of the data	A random variable $\hat{V}(x_0) = \Sigma\, w_i \cdot V(x_i)$, whose parameters are known because it is a linear combination of known random variables
Error	A specific but unknown quantity $r_0 = \hat{v}_0 - v_0$	A random variable $R(x_0) = \hat{V}(x_0) - V(x_0)$, whose parameters are known for the same reason
Problem	How to weight the samples so that the average error is 0	How to weight the random variables so that the expected value of the estimation error is 0

The key idea: we cannot control the actual error at one location, but within the random function model we can choose weights so that the error is unbiased on average (and, further, so that its variance is minimized, which leads to kriging).

Estimation and Multivariate Relationships

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Multivariate relationships are often related to collocated observations of different variables. However, they can also relate to variables that are sampled at different locations but in the same domain.

Equally sampled

The relationship is not really a constraint that affects the conditional distribution/uncertainty. For estimation, using cokriging results in screening of the secondary variables. Thus, for estimation of equally sampled data, each variable is estimated separately. This makes sense, since the estimates are not expected to honor any kind of multivariate relationship.

For simulation, we use PPMT, hierarchical simulation, or cokriging for unconditional simulation followed by CBK (conditioning by kriging). This is because in simulation we draw values of the variables at unsampled locations, not moments. Thus, simulation is expected to reproduce the multivariate relationships, as they can affect post-processing transfer functions.

Sampled at completely different locations

For both simulation and estimation, we fit an LMC and use a linear measure of relationship (i.e. covariance) from the direct and cross variograms to get the conditional expected value (i.e. the estimate) or to quantify the conditional distribution for simulation. This is because different variables available at different locations can contribute to the conditional uncertainty or the estimate.

Factors affecting Uncertainty

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

The factors mentioned below affect the local uncertainty at different scales. They are also important for global uncertainty, but here the focus is to understand local uncertainty vs data spacing at different scales.

Major ones

- **Global histogram**
  - The overall uncertainty
  - Proportional effects
- **Variogram at short scales**
  - Gamma bar: $D^2(v, V) = D^2(\cdot, V) - D^2(\cdot, v)$ can give us an understanding of the prior variance reduction after upscaling (before conditioning).
  - The effect on conditioning.
  - Variogram range compared to domain size can affect global uncertainty: more continuous → larger global variability, and less local variability.
- **Conditioning by data**
  - Data spacing
  - Variogram
- **Clipping by domain**

Minor

- **Trend**
  - Large-scale variability
  - Contribution to block/panel upscaling
  - Proportional effect
- **Non-Gaussianity in conditioning data**
- **Contacts**

Gibbs Sampling

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Gibbs sampler is a Markov Chain Monte Carlo (MCMC) algorithm for dependent sampling from a multivariate distribution function. It works for 2+ dimensions.

Summary of the clipped reference text on the page: Gibbs sampling draws a sequence of samples from a multivariate distribution when direct sampling is difficult but sampling from each conditional distribution is easy. Starting from an initial value, each component is sampled in turn from its distribution conditional on the current values of all other components, and the step is repeated. The samples approximate the joint distribution; early samples (burn-in) are usually discarded, and successive samples are correlated, so thinning is sometimes used.

> The conditionals are known, rather than the marginals.

Consider a 3D example

Figure 1. Axes x_1, x_2, x_3.

$(x_1^0, x_2^0, x_3^0) \leftarrow$ initial point

First sample:

$x_1^1 \leftarrow f(x_1 \mid x_2 = x_2^0, x_3 = x_3^0)$

$x_2^1 \leftarrow f(x_2 \mid x_1 = x_1^1, x_3 = x_3^0)$

$x_3^1 \leftarrow f(x_3 \mid x_1 = x_1^1, x_2 = x_2^1)$

$\Rightarrow (x_1^1, x_2^1, x_3^1)$

Second sample:

$x_1^2 \leftarrow f(x_1 \mid x_2 = x_2^1, x_3 = x_3^1)$

$x_2^2 \leftarrow f(x_2 \mid x_1 = x_1^2, x_3 = x_3^1)$

$x_3^2 \leftarrow f(x_3 \mid x_1 = x_1^2, x_2 = x_2^2)$

$\Rightarrow (x_1^2, x_2^2, x_3^2)$

Heteroscedasticity and Heterogeneity

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Heterogeneity: refers to the variability of a geological variable in a certain volumetric support. In other words, the spatial variation of a geological variable. More spatial continuity results in lower heterogeneity and smoother-looking models.

Heteroscedasticity: when the variability of a geological variable (e.g. grade) is different at different locations. In other words, second-order non-stationarity. When the variability and mean are correlated, it is referred to as the proportional effect.

In general, both types of non-stationarity can affect the accuracy of estimates. For simulations, a normal score transformation followed by trend modeling helps to mitigate the proportional effect/heteroscedasticity and first-order non-stationarity.

> Note: it is common to use moving average statistics to assess relationships between the local average and variability.

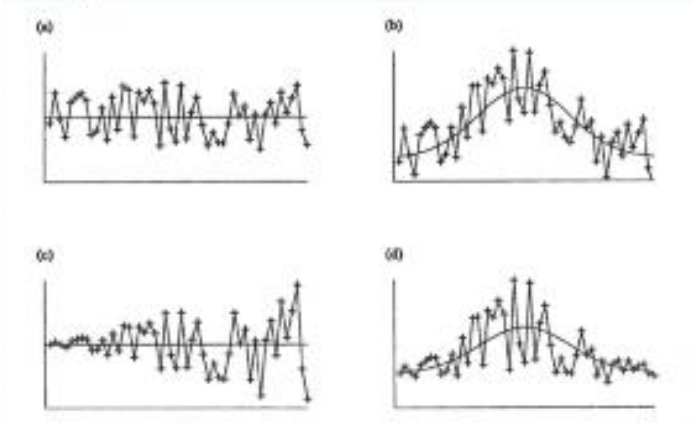


Figure 1. Reference figure pasted on the page: hypothetical data profiles showing (a) constant local mean and variability, (b) a trend in the mean only, (c) a trend in the variability only, and (d) trends in both.

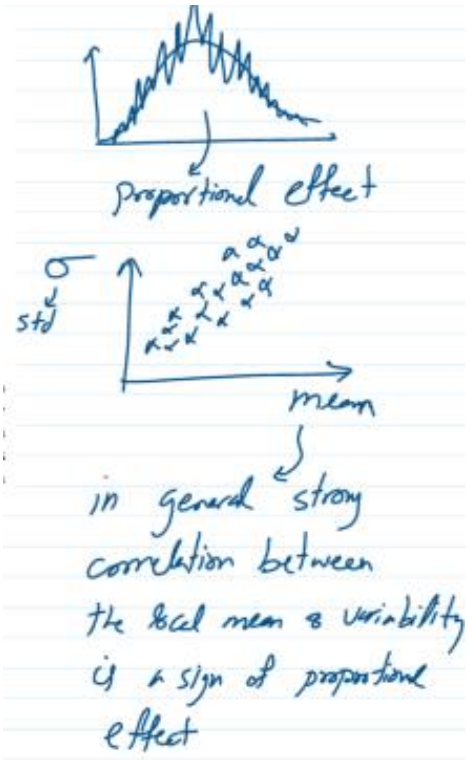


Figure 2. Proportional effect: local standard deviation increasing with the local mean.

In general, a strong correlation between the local mean and variability is a sign of the proportional effect.

Importance of Nugget in Estimation

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

The nugget refers to very small scale variability. In practice, the nugget depends on the support volume at which the composites/data are available.

Intuition

1) Consider the composites are available at 1 m. In practice, this means the support volume at which we can develop the highest resolution model is 1 m^3. The nugget is the variability between two 1 m^3 blocks of rock sampled/extracted infinitely close to each other.

2) If the goal is to model at the same scale as the composites, then the nugget is more of a fitting problem than a contributing factor to the estimation/simulation.

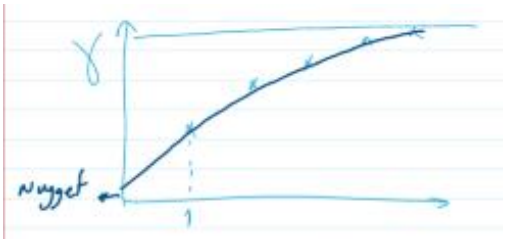


Figure 1. Variogram with a nugget effect at the origin.

For grid estimation, most of the time the distance between blocks and composites is more than 1 m, except when the composite centroid is inside a block. Therefore, the nugget is not that important.

3) For change of support, where the estimate is at a larger block (block kriging), the $\bar{\gamma}$ (gamma bar) and volume variance are affected by the nugget.

Interpretation of Cross Validation Plots

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

The cross validation plots compare the estimate (single value) to the actual data (usually done for OK). There are different ways to do this, including:

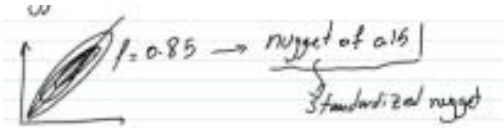
- Jackknife
- Leave one sample out (more common)
- Leave one drill hole out (more common)

Leave one sample out

When leaving one sample out, depending on the orientation of the drill holes, different interpretations are possible.

1) If wells are vertical, then the vertical variogram has the dominant effect on the result of leave-one-sample-out.

2) One rule of thumb with OK is that after removing one sample, the samples above and below have strong control due to screening effects, and since they share similar (identical) covariance with the missing location, one can argue that the estimate is close to the average of the two samples above and below. Therefore, the correlation reduction from 1.0 can be used to infer/assess the nugget.



e.g. ρ = 0.85 → nugget of 0.15 (standardized nugget)

Kriging Compare to Other estimation techniques

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Kriging vs "IDW (power of distance), NN (polygonal), triangulation, simple average"

Similarities

- Linear combination of random variables.
- Global unbiasedness: $\Sigma \lambda_i = 1$.
- Smoother results with more data.

Differences

- Kriging minimizes the error variance.
- Kriging can result in negative weights and produce estimates that are higher or smaller than the data.
- Kriging uses statistical distance through the covariance and can account for data redundancy, so there is less adverse effect of clustered data.
- Kriging can be used for direct block kriging with consistent results.

LCS vs E-type

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Localized Conditional Simulation

LCS is based on using all realizations to create a single model that has realistic variability/heterogeneity like a realization and reproduces average realization statistics.

E-type	Localized
Conditionally unbiased (using a lot of data)	Conditionally biased
Smooth	Realistic variability (reproducing data)

A localized model can be used for open pit and design of the pit extent but, for underground, which has a higher level of selectivity, the conditional bias can be misleading for designing the stopes.

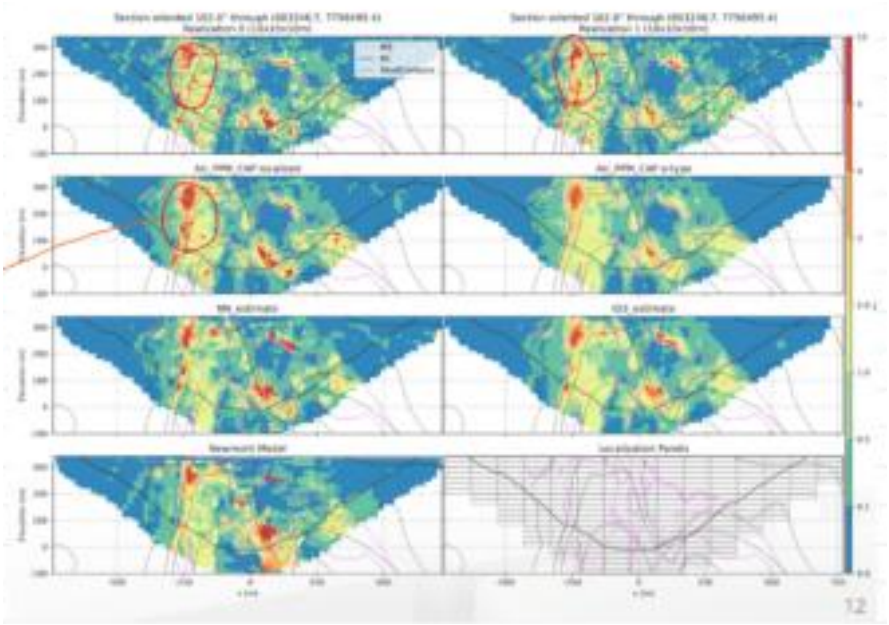


Figure 1. Sections through realizations, estimates and the localized model. Note how the localized model has higher values in high locations compared to the realizations.

This could result in conditional bias, meaning that at high locations the LCS estimate tends to be higher than average.

Panel size:

- Bigger → moving grade around more.
- Smaller → higher expression of conditional bias.

> Note that conditional bias is a matter of local bias. The LCS is not globally biased, as it follows the average of the realizations.

PDF, Mean, Variance, covariance

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

A probability density function (PDF) is a density, since it gives probability per data extent.

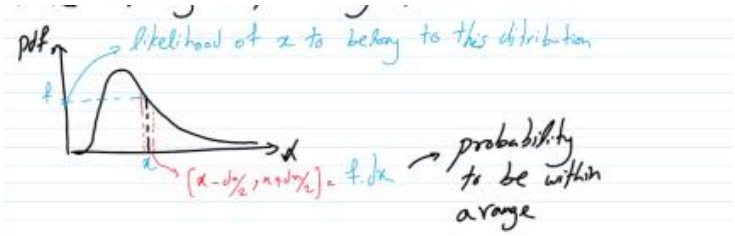


Figure 1. f(x) is the likelihood of x belonging to the distribution; the probability of being within the range [x − dx/2, x + dx/2] is f·dx.

The **mean** is the first moment and can be described as the "centre of mass", since its calculation is similar to the centre of mass calculation:

$$\bar{x} = (1/n)\ \Sigma_{i=1..n}\ x_i$$

But to infer the mean from a distribution:

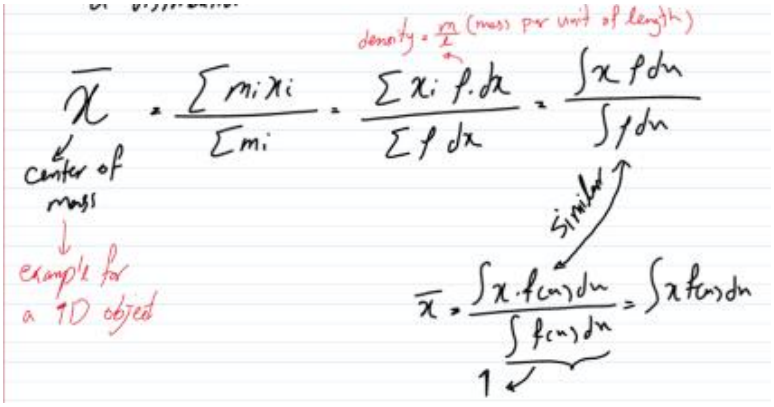
$$\bar{x} = \int_{-\infty..\infty} x \cdot f(x)\ dx$$

Centre of mass, for example for a 1D object, where density ρ = m/L (mass per unit of length):

$$\bar{x} = \Sigma\ m_i x_i\ /\ \Sigma\ m_i = \Sigma\ x_i \cdot \rho \cdot dx\ /\ \Sigma\ \rho\ dx = \int x\ \rho\ dx\ /\ \int \rho\ dx$$

Similarly, for a distribution:

$$\bar{x} = \int x \cdot f(x)\ dx\ /\ \int f(x)\ dx = \int x\ f(x)\ dx \text{ (since } \int f(x)\ dx = 1)$$



Point Scale to SMU Scale

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

A point scale grid should not be regarded as block volumes; instead, it should be considered a discretization to get enough point scale values inside each SMU block to support block averaging and change of support.

In this context, we often choose the point scale grid size in a way that the resolution of our vertical simulation matches the resolution of our conditioning data. The horizontal resolution is chosen to provide enough point scale values to support averaging within the SMU block. It is best to think of simulated nodes rather than blocks, that is, simulating variability at point scale and at different locations.

For example, with 2 m composites:

> point scale: 2 × 2 × 2; SMU scale: 8 × 8 × 4 → we will get 32 (4 × 4 × 2) point scale samples within an SMU block

So, the decision is driven by:

1) Composite length and matching the vertical resolution for point simulation. For underground mining with directional drilling, this could get more complicated.

> Note: the core diameter is so small that each composite is a point scale with a certain length.

2) The horizontal and vertical resolution of the point scale grid results in enough point scale simulated nodes within each SMU block to support an accurate change of support.

3) Trade-off between accuracy and computational cost/time.

After upscaling to the SMU block, we can assign grade values to a block and calculate resources.

Proportional Effect and Data Spacing

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Definition

When local variability is not stationary and depends on the local mean. Thus, it can be regarded as second-order non-stationarity.

Intuition regarding data spacing

With fixed/regular data spacing, the proportional effect does not exist in Gaussian units. After the NS transform and trend removal, the distribution of the data is Gaussian, so the map of local uncertainty (let's say the standard deviation) is uniform.

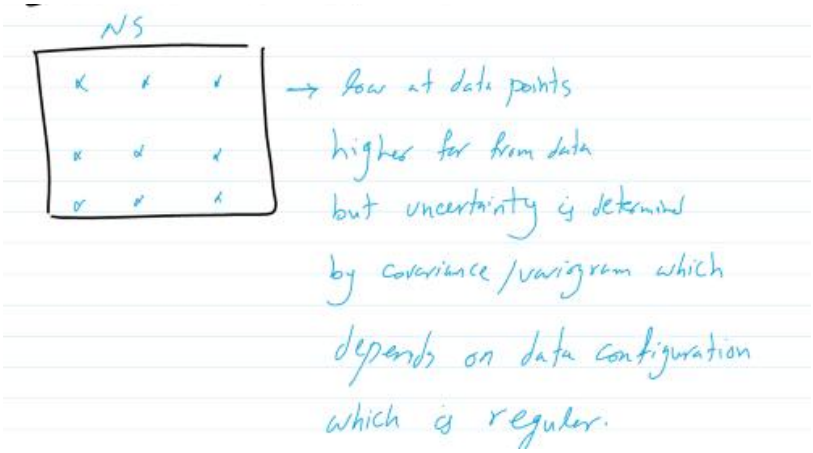


Figure 1. Regular data grid in NS units: uncertainty is low at data points and higher away from data, but it is determined by the covariance/variogram, which depends on the data configuration, which is regular.

After reintroducing the trend through the SCT back-transform, it is possible to observe some proportional effects, since the relation between the data and the collocated trend can be heteroscedastic.

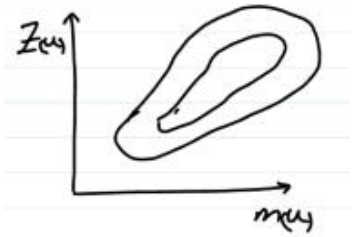


Figure 2. Heteroscedastic relation between Z(u) and m(u).

Finally, after back-transformation to original units, the skewness of the original distribution results in stronger proportional effects, e.g. higher variability at locations with higher values.

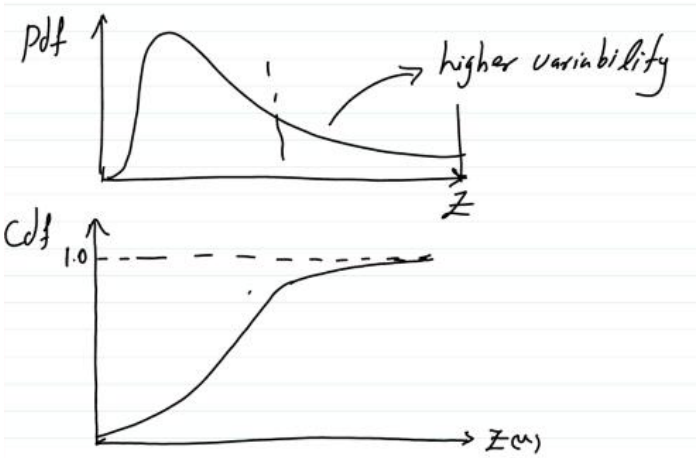


Figure 3. Skewed PDF and corresponding CDF of Z(u): the long upper tail has higher variability.

RMSP Trend removal steps

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

1) NS transform fit based on the provided trend model (estimation) or the collocated trend values at data locations. Save the fitted NS transformer.

> Note: the user can provide a trend variable being a column in both the composites and the trend model. This allows the trend removal transformation of composites outside the grid. A user should arrange for the trend calculation at composite locations to benefit from this option.

2) Fit a GMM between collocated values that are usually NS transformed and the trend value. Note that an NS transform of the variable before trend modeling helps with a better GMM fit.

3) Implement SCT for the variable conditional to its collocated trend value, using the conditional distribution or CDF inferred by the GMM. Store the SCT transformer.

Now that the transformers are fitted and saved, one can pass the variable and trend model to do the forward transformation (de-trending).

Note that the forward transformation for the trend model is just an NS transform using the already fitted NS transformer.

Search Anisotropy and Variogram Search

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

For a variogram-based estimation, the search anisotropy is expected to match the variogram anisotropy, or be bigger and inclusive in all directions.

The search anisotropy can affect which data are included in the estimation process, but it does not affect the kriging weights, as they are informed by the variogram model and its anisotropy.

However, a small search can result in search artifacts. When a search has larger ranges than the variogram in all directions, its anisotropy has no effect on the estimation.

Trend model

For a trend model, the search anisotropy controls the final model anisotropy, since distances are scaled by the search range before calculating the kernel weight. This is true when the bandwidth of the kernel is defined based on the search ranges.

NN estimator

For the nearest-neighbour (NN) estimator, the search anisotropy controls which point is closer. For example, with a horizontal/vertical anisotropy of 1000, the horizontal distances become very dominant in determining the closest point. This can be used for declustering weight calculation when there is no clustering along the vertical direction.

Intuition

When the search ranges are included in a spatial weighting system, like what happens with a moving average trend model, the ranges affect the spatial weighting.

e.g. search ranges: 5000, 5000, 25 (domain 200 × 200 × 50 m)

→ This results in a vertical trend, since the huge anisotropy means points on the horizontal plane get more or less the same weight, due to the huge ranges compared to the domain size. Also, the horizontal-to-vertical anisotropy results in significantly smaller weights for points at a vertical distance.

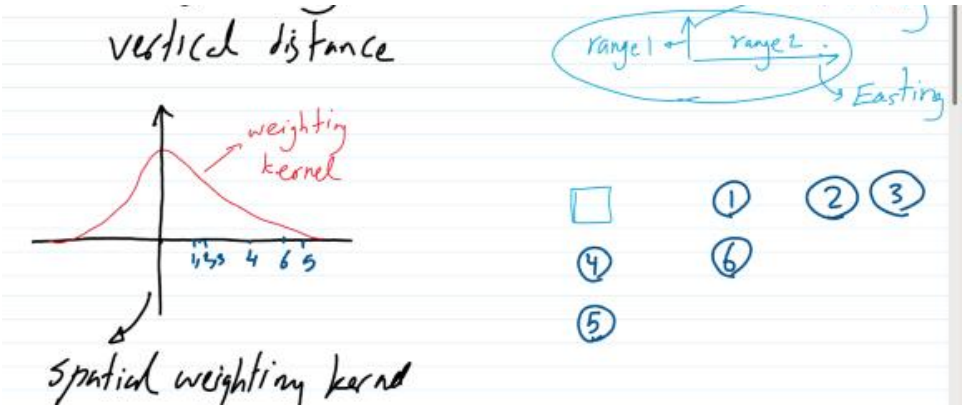
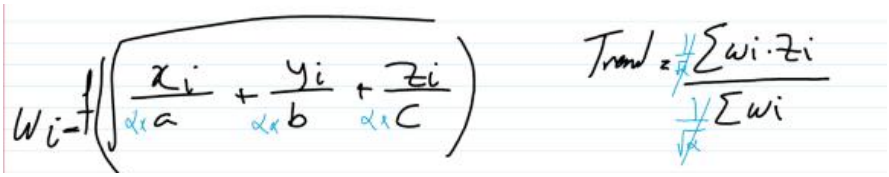


Figure 1. Spatial weighting kernel (left) and the search ellipse with range 1 and range 2 along Northing/Easting, with numbered data locations (right).

Also, note that while keeping the anisotropy, scaling all the trend search ranges has no effect on the trend, as long as the same number of data locations are included:



$w_i = f\left(\sqrt{ x_i/(\alpha \cdot a) + y_i/(\alpha \cdot b) + z_i/(\alpha \cdot c) } \right)$, $\text{Trend} = \Sigma\, w_i \cdot z_i \,/\, \Sigma\, w_i$ (the scaling factor α cancels out)

Spatial Bootstrap and Conditioning Data

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

Background

The spatial bootstrap (SBS) is designed to quantify histogram uncertainty given a reference data distribution and a model of spatial continuity (i.e. a variogram model). SBS realizations are generated by unconditional simulation at data locations.

Variogram and conditioning data

Including histogram uncertainty is expected to affect (increase) the overall global uncertainty. In this context, the effect of variogram continuity is two-fold:

1) More continuity means more dependency between the data, and thus the number of effective data is lower. So, with a continuous variogram, the SBS realizations at data locations have more uncertainty. This makes sense because the unconditional simulation at data locations will have more global uncertainty as the variogram range increases (i.e. more continuity).

2) The second fold is that the variogram continuity increases the conditioning effect of the data. In the presence of dense data, the second role can be more dominant, as the histogram of a realization is controlled by the conditioning data values rather than the reference SBS distribution that is used for the transformation (usually the NS transform).

Spike and NS transform

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

A spiky distribution refers to a situation where a single value is repeated or has many incidences of very close values.

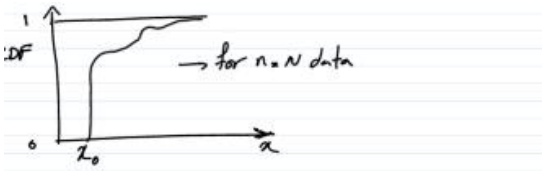


Figure 1. CDF with a spike at z_0 (for n = N data).

The problem is that during the fitting phase, a map between original values and normal values is established. Then, transformation is done using the map as a lookup table:

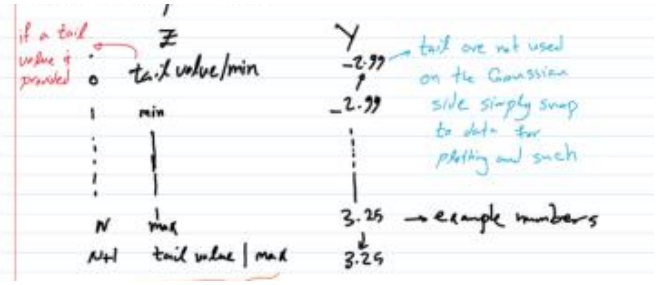


Figure 2. Lookup table between Z and Y. If a tail value is provided, it is added at index 0 (tail value/min) and N+1 (tail value/max).

	Z	Y
0	tail value / min	−2.99
1	min	−2.99
⋮	⋮	⋮
N	max	3.25
N+1	tail value / max	3.25 (example numbers)

> The tail values are not used on the Gaussian side; they simply swap in to show for plotting and such.

Forward transformation: data beyond the tail values may raise an error unless the option "clip_by_ref" is used. Then, the data are clipped by the tail values.

Back transformation: any value beyond the normal lookups will be extrapolated to match between max and upper tail, or min and lower tail. No exception is raised ever.

Now, the spike problem is when the same values, i.e. z_0, are being looked up. In such cases, we end up with the same normal values for all observations of z_0. Depending on the amount of spikiness,

different levels of despiking should be implemented prior to the normal score transformation.

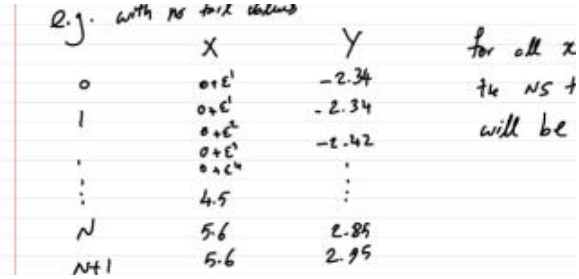


Figure 3. Example with no tail values.

	X	Y
0	$0 + \varepsilon^1$	−2.34
1	$0 + \varepsilon^2$	−2.34
⋮	$0 + \varepsilon^3$, $0 + \varepsilon^4$	−2.42
⋮	4.5	⋮
N	5.6	2.85
N+1	5.6	2.95

For all X values = 0, the NS-transformed value will be −2.34.

> Note: ε values are added at machine precision to make sure the lookup table is stable and has a unique response.

Super Secondary

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

A super secondary variable is an aggregation of multiple secondary (often exhaustive) variables into one representative variable.

RMSP standardizes the primary variable, as it asks for the weight too. The secondary variables are left to users to be standardized at the exhaustive level, then resampled at data locations.

The final super secondary variable is a linear combination of all secondary variables:

$$SS = \Sigma_{i=1..n} \mu_i S_i$$

The final SS variable is often further standardized at the exhaustive level before being used for estimation.

For cokriging, and often ICCK, the estimator needs to know the:

- SK mean → mean of the primary
- std of the primary
- mean of the secondary
- std of the secondary

⇒ These are used to standardize the primary and secondary variables involved in cokriging.

Note: for kriging (one variable), there is no need to standardize variables, but the variogram is often standardized, which is not important, as it is used as a relative measure of spatial variability/continuity per Euclidean distance.

Note: in the simulation context, the NS transform is performed for the primary variable and the exhaustive secondary, and there is no need to worry about the standardization.

Note: with secondary data, the exhaustive data are transformed (NS/standardized) before sampling at drill locations. One main advantage of using secondary or super secondary data is debiasing the estimate/simulation of the primary variable. It is possible that the sampled secondary variable at data locations does not cover the full range of variability and other statistical measures of the secondary variable(s). Thus, cokriging or ICCK can debias the estimation of the primary variable in case the sampled data are biased.

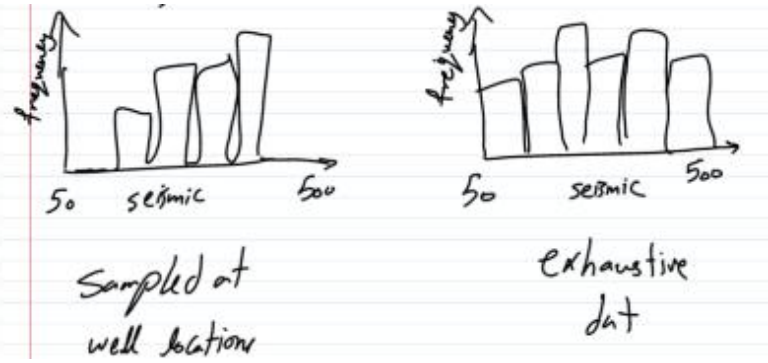


Figure 1. Histogram of the seismic attribute sampled at well locations (left) versus the exhaustive data (right).

Trend Intuition

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

A trend model is a smooth, consistent map or model that can often be modelled deterministically.

In the context of simulation, we are interested in quantifying the conditional distribution of a random variable at an unsampled location, conditioned to nearby data:

$$Z^*(u_0) - m(u_0) = \Sigma_{\alpha=1..n} \lambda_\alpha \, (Z(u_\alpha) - m(u_\alpha))$$

$$\sigma^2_{SK} = \sigma^2_{conditional} = C_0 - \Sigma_{\alpha=1..n} \lambda_\alpha \, C(u_\alpha, u_0)$$

In the probabilistic context, $m(u_\alpha)$ and $m(u_0)$ are the mean values (locations of) the marginal distributions of the random variables $Z(u_\alpha)$, $Z(u_0)$.

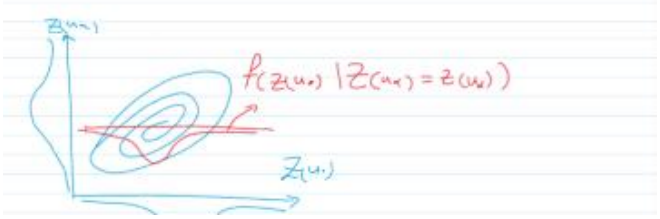


Figure 1. Conditional distribution f($Z(u_0)$ | $Z(u_\alpha) = z(u_\alpha)$).

Note that in the Gaussian framework, every location is a random variable, and we assume that there is a multivariate Gaussian distribution between all locations that can be quantified by a vector of means and a matrix of covariances between pairs of variables.

So, the trend model is the vector of marginal means. Considering an ensemble of 100 conditional realizations, the 100 values of each grid cell quantify the conditional distribution at that location, given the conditioning data and sources of uncertainty.

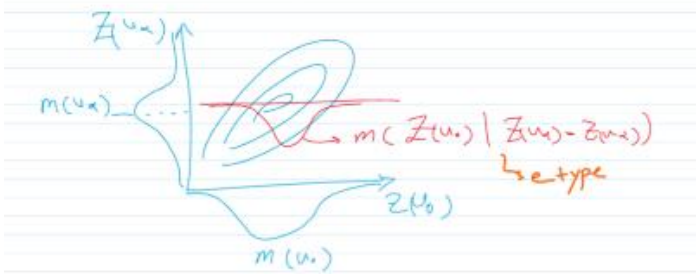


Figure 2. Marginal mean $m(u_\alpha)$ versus the conditional mean m($Z(u_0)$ | $Z(u_\alpha) = z(u_\alpha)$), i.e. the e-type.

In theory, the covariance of two random variables should be the same before and after trend removal. However, the problem is that the following relationship between the variogram γ(h) and the covariance function is valid only under the stationarity assumption:

$$\gamma(h) = C(0) - C(h)$$

Proof:

$$\gamma(h) = \tfrac{1}{2} E\{ [Z(u) - Z(u+h)]^2 \}$$

$= ½ (E[Z^2(u)] - 2E\{Z(u) \cdot Z(u+h)\} + E\{Z^2(u+h)\})$

$= ½ (E[Z^2(u)] - m^2(u) - (2E\{Z(u) \cdot Z(u+h)\} - 2m(u)m(u+h)) + E\{Z^2(u+h)\} - m^2(u+h) + m^2(u) + m^2(u+h) - 2m(u)m(u+h))$

$\gamma(h) = ½ (\sigma^2(u) + \sigma^2(u+h)) - C(u, u+h) + ½ (m(u) - m(u+h))^2$

With first- and second-order stationarity, $m(u) = m(u+h)$ and $\sigma^2(u) = \sigma^2(u+h)$:

$\Rightarrow \gamma(h) = C(0) - C(h)$

That is why trend removal, and variogram calculation and modeling for the residuals, is considered.

Intuition point 1

In the case of unconditional simulation, the ensemble of L (e.g. 100) realizations for each point/cell provides the marginal distribution of the target random variable at that location (i.e. the centre of the cell).

If the trend removal is a simple residual ($R = Z(u) - m(u)$), then the e-type and the trend model are the same in theory.

Intuition point 2

If the trend removal is based on SCT between the trend and the target variable at those locations, then:

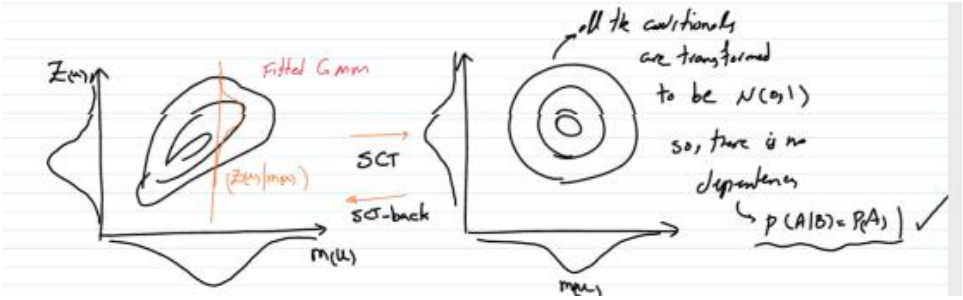


Figure 3. Fitted GMM between Z(u) and m(u); after SCT, all the conditionals are transformed to N(0,1), so there is no dependence: P(A|B) = P(A).

Clayton thinks that with unconditional simulation, the e-type will be matched with the mean of the conditional distribution from the fitted GMM. Mathematically, it seems that he is correct, because all the realization values at a location are transformed back to honor the relationship with the trend characterized by the GMM.

So, e-type = $E\{ Z(u) \mid m(u) \}$

The concept of SCT + GMM is not intuitively integrated with random variables and the marginal distribution.

Trend Removal

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

The idea of trend removal is that an aspect of the domain can be modeled deterministically based on available data. This deterministic aspect is the trend and can be removed, and the remaining stochastic/uncertain aspect can be modeled separately:

- The residual can be simulated and all realizations are added to the shared deterministic trend.
- Calculating the residual is the conventional way, and it is expected that the residuals are not correlated with the trend.
- However, the current approach is to use SCT + GMM to efficiently decorrelate the trend and collocated data, and hope that it also makes the transformed/de-trended data stationary.
- One other approach is to calculate the residual and normal score (NS) it; then, if there is any correlation with the trend, consider SCT + GMM.

A good trend model is key, and it is tricky to evaluate:

- NS variograms before trend removal and after de-trending are supposed to be very close at small scale.
- After de-trending, the continuity is expected to be reduced. Also, it is expected that variograms become well-behaved as the non-stationary features are removed.
- If the trend removal is done properly, the variogram reaches the sill and remains around it.

Anisotropy

If there is a geometric anisotropy, like a huge $Range_{horiz}/Range_{vert}$, the trend (deterministic aspect) is expected to respect it.

The search range anisotropy should be inferred from the variogram, especially horizontal/vertical. This is true for trend modeling tools that infer the weighting kernel bandwidth from the search range (e.g. RMSP):

- Anisotropy directions: inferred from the variogram.
- Anisotropy ratio: inferred from the variogram. The trend is supposed to produce a portion of the anisotropy; the rest is handled by the residual simulation.

Trend model parameters

As mentioned above, the anisotropy of the trend will be inferred from variograms (e.g. NS variograms). This is often inflated by variogram buffers, and while the anisotropy is maintained, the ranges are scaled to make sure all grid cells within the domain will have a trend model.

The other important parameter is the maximum number of points within the search ellipsoid. The trend model is expected to be smooth enough; otherwise it takes all the variability away. One approach is to calculate the trend model at data locations with a search template, change the

maximum number of composites, and monitor the residual and trend correlations.

An optimal trend is expected to minimize the residual correlation, meaning that the residuals do not depend on the trend.

Trend model, local uncertainty and spatial continuity

Overfitting the trend model:

- Can result in understating local uncertainty.
- Effect on spatial continuity? (open question in the notes)

Underfitting the trend model has the reverse effects.

Variogram Non-Stationary

OneNote notebook RMS / section Geostat. Transcribed from handwritten notes.

In general, a variogram can be calculated for any two variables with paired observations:

$$2\gamma_{xy} = \Sigma_{i=1..n} (x_i - y_i)^2$$

$$2\gamma_{xy} = \sigma_x^2 + \sigma_y^2 + (m_x - m_y)^2 - 2\sigma_{xy}$$



Figure 1. Paired observations of x and y.

Extending to a spatial random variable:

$$2\gamma(h) = 2C(0) + (m(u) - m(u+h))^2 - 2C(h)$$

- $2C(0)$ becomes $(\sigma^2(u) + \sigma^2(u+h))$ if there is no second-order stationarity.
- $(m(u) - m(u+h))^2 = 0$ if stationary.

As you can see, in the presence of a strong trend, the difference between the mean values (m(u), m(u+h)) can increase the variogram value and result in observations above the sill.



Figure 2. Experimental variogram points rising above the sill of 1.0.

Notes

- The fact that the variogram value only depends on the lag vector magnitude and orientation, and not the tail location, is due to the second-order stationarity assumption.
- The symmetry of the variogram is related to its mathematical derivation:

  $\gamma(h) = \gamma(-h)$

  $(z_i - z_j)^2 = (z_j - z_i)^2$ (direct)

  $(z_i - z_j)(y_i - y_j) = (z_j - z_i)(y_j - y_i)$ (cross)

Chapter 3

Mining

20 sections

Angle Conventions (Leapfrog Support)

OneNote notebook RMS / section Mining. The page holds a pasted Seequent support article ("Angle Conventions in Leapfrog Edge"); this is a summary of it, not a copy.

- Leapfrog Edge uses the same angle convention as Leapfrog: a direction in space is given by Dip, Dip Azimuth and Pitch in the geological sense.
- Leapfrog uses a right-hand rule throughout, based on the geologist's plane. The rotation order is ZXZ: first around Z by the dip azimuth, then around the new X by the dip, and finally around the new Z by the pitch. All rotations are clockwise when looking along the positive axis (so the first rotation is clockwise when looking down).
- Datamine rotations are the same as Leapfrog's. Isatis supports multiple conventions; the one that matches Leapfrog was reported as unreliable, so using the Datamine-equivalent convention is recommended.
- Gemcom (as used through Supervisor) can also use a ZXZ rotation, but some angles need to be inverted.
- Another common convention is the GSLIB one: a ZXY rotation order with the final Y axis as the major axis, whereas Leapfrog uses the final X axis. Even though the two look similar for general rotation conversion formulas, the differences need to be accounted for. Surpac and Vulcan can use the GSLIB-style convention, although the last rotation is in the opposite direction.

In each case, check the resulting orientation and axes in 3D.

Assay Types

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

1) Fire assay: the industry standard for obtaining gold and platinum group element (PGE) concentrations from high grade ores.

2) Shake leach: a quick, optimistic analysis to determine the amount of cyanide-extractable gold. If the gold is cyanide (a chemical solution) soluble, the shake leach and fire assays will be highly correlated.

> Note: the shake leach grade should be equal to or less than the fire assay.

Compositing

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Drill hole data consist of:

- **Collars:** location of the drill hole header (x, y, z), where z is the elevation.
- **Assay:** interval data along the drill hole with regular/irregular interval lengths; From and To give the depth along the hole.
- **Survey:** the orientation of drill hole line segments; azimuth and inclination as a function of depth along the hole.

The process of compositing is to aggregate the interval/assay data into larger and more regular length segments.

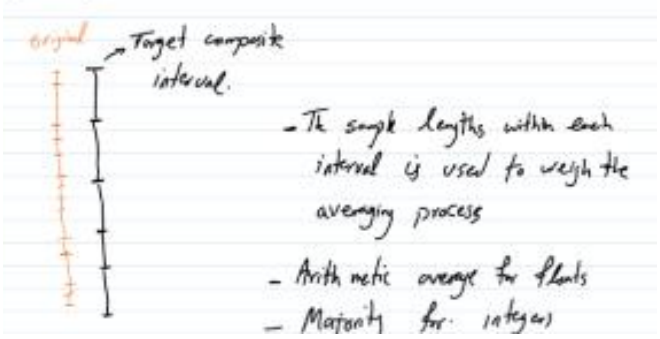


Figure 1. Original sample intervals versus the target composite intervals.

- The sample lengths within each interval are used to weight the averaging process.
- Arithmetic average for floats.
- Majority for integers.

An indicator of 0/1 with float type will be converted into a proportion.

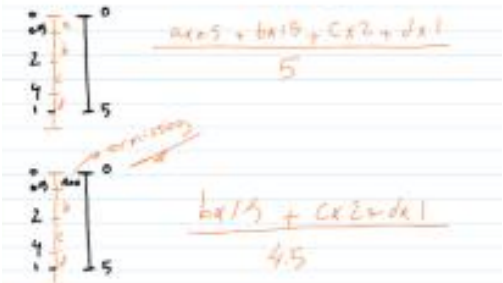


Figure 2. Example: composite 0-5 m from intervals a, b, c, d with lengths 0.5, 1.5, 2 and 1: (a×0.5 + b×1.5 + c×2 + d×1) / 5. If interval a is missing: (b×1.5 + c×2 + d×1) / 4.5.

As shown above, the composited value is the length-weighted arithmetic average. Categoricals are based on majority rules with length-weighted averaging. Missing intervals or missing/NaN values are not included in the averaging.

Hint: if it is required to calculate the length of a category, one solution is to fill the missing gaps and make sure the drill hole is contiguous. Then, get the proportion of the category using a binary indicator and arithmetic averaging. Then multiply that proportion by the final composited length for each interval.

Cutoff grade and curves

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

The cut-off grade is determined based on economics, to determine if it is profitable to process a rock block or SMU.

Based on the cut-off, a model is divided into ore and waste/non-ore parts. The average metal grade for grid cells above the cut-off is the average ore grade, and the total tonnage of block cells above it is called the ore tonnage.

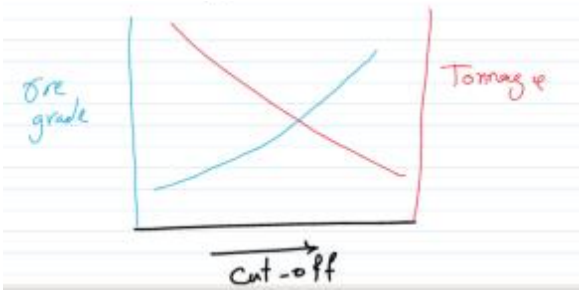


Figure 1. Grade-tonnage curves: ore grade increases and tonnage decreases with increasing cut-off.

grade: ppm ≅ g/ton

tonnage: $\Sigma V_{cell} \times SG$ (consider the cut-off grade filter for the calculations)

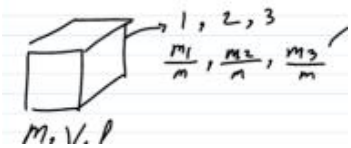
Σ metal grams = ore tonnage · average ore grade

Note: the SMU scale is the smallest mining unit for decision making and determining ore/waste based on a grade or NSR (net smelter return).

Density Calculations

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

We have weight/mass percentages m_1/m, m_2/m, m_3/m for three variables/constituents in a block with mass M, volume V and density ρ.



Finding the total density based on the individual densities and weight percentages:

$\rho = m / V$

$V = V_1 + V_2 + V_3 \Rightarrow m/\rho = m_1/\rho_1 + m_2/\rho_2 + m_3/\rho_3$

$1/\rho = (m_1/m)/\rho_1 + (m_2/m)/\rho_2 + (m_3/m)/\rho_3$

$1/\rho = w_1/\rho_1 + w_2/\rho_2 + w_3/\rho_3$ (w_i: weight fraction)

total mass = ρ · V

Dilution and Volume Variance Correction

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Volume variance correction only accounts for the "internal dilution", based on the correction factor that is informed by the variogram.

Types of dilution:

- Internal dilution
- Geologic contact dilution → sub-grids and domain definition
- Operational mining dilution

Drilling

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Diamond drilling (DD) and reverse circulation (RC) are the two main established drilling technologies.

- "Diamond core drilling" is good for precise sampling and analysis of layers.
- "RC" is good for extracting bulk samples.

In general, RC is quite a bit faster than DD.

Engineering Terms and Units

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Dilution (%) = tonnage of waste milled / (tonnage of ore milled + tonnage of waste milled)

Recovery (%) = extracted ore / (extracted ore + unextracted ore) × 100

Grade units

ppm and g/ton are the usual units. If the ton is metric, ppm is identical to g/tonne.

Comminution → the action of reducing a mineral or an ore to minute particles or fragments.

Estimation and Conditional Bias

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Conditional unbias

For an estimator (e.g. SK or OK), conditional unbiasedness is different from global unbiasedness. With Z*(u) the estimate:

> Global unbiasedness: E{Z*} = E{Z} (the expected value of the estimates matches the expected value of the random variable)

For both SK and OK, global unbiasedness is satisfied by design or constraints.

> Conditional unbiasedness: E{ Z | Z* = z } = z

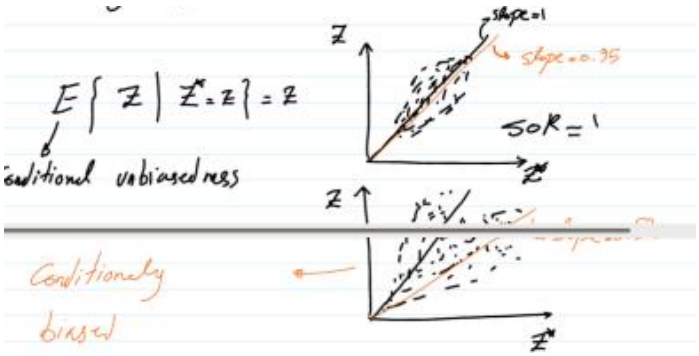


Figure 1. Top: slope of regression (SoR) ≈ 1 (e.g. 0.95), conditionally unbiased. Bottom: slope ≈ 0.52, conditionally biased.

In other words, the slope of regression between Z* (feature) and Z (label) should be 1 in order to have conditional unbiasedness.

SK is similar to a classical linear regressor that minimizes squared error. Thus, it is conditionally unbiased by design.

> $y = ax + b \rightarrow a = C_{xy} / C_{xx}$ (the SoR for simple linear regression), $b = E\{y\} - a\,E\{x\}$
>
> $Z = aZ^* + b$, $a = C(Z, Z^*) / \mathrm{Var}\{Z^*\}$, $b = E\{Z\} - a\,E\{Z^*\} = (1 - a)\,Z$

For SK it can be shown that:

> $\mathrm{Var}\{Z^*\} = \Sigma_{\alpha=1..n}\, \lambda_\alpha\, C(u_\alpha, u_0)$
>
> $\mathrm{Cov}\{Z, Z^*\} = \Sigma_{\alpha=1..n}\, \lambda_\alpha\, C(u_\alpha, u_0) \rightarrow$ thus a = SoR = 1 and b = 0

The key for SK to be conditionally unbiased is its system of equations: $\Sigma_{\beta=1..n}\, \lambda_\beta\, C(u_\alpha, u_\beta) = C(u_\alpha, u_0)$.

For OK, the stationarity is defined locally and within a search. The search radius and the number of composites or conditioning data determine the level of conditional biasedness for OK.

- **Large search** → reduces the conditional biasedness at the expense of lower variance for the estimate, i.e. smoothness.
- **Limited search** → good variance reproduction at the expense of being conditionally biased.

Note: an intuition of conditional biasedness is that the estimator estimates high values to be even higher and low values to be lower.

Estimation Models

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

"The most important goal of estimation is to predict future grade and tonnage that can be mined."

Long Term Model (LTM)

- Long term model → widely spaced drill holes.
- Final/grade control model → blast holes and stope grade data.

LTM:

- **Open pit:** local precision of estimates is not a priority. The emphasis is on the accuracy of the estimated global recoverable resources.
- **Underground:** local precision and accuracy matter.

Production model

The production model is also referred to as the ore/grade control model and has the following characteristics:

- Built from blast hole/production data.
- Used for the final ore/waste decision.
- Used for reconciliation against long term or resource models.
- Limited to the extent of the data.
- Data often available in large numbers and regularly spaced:
  - Blast holes → horizontal grid and bench height.
  - One sample per drill hole.
  - Lower quality data compared to DDH/RC.
  - Variogram calculation is easier to automate.
- Considering the high density of data, the production model is practically assigning the data to the grid (depending on grid size). Therefore, an ID estimate often provides results similar to OK.

> Note: a comparison between the long term/resource model and the production model is often done to assess reconciliation. Disseminated deposits usually reconcile better than structurally controlled deposits.

Kriging search: the search range is often limited for the production model to avoid extrapolation beyond the production data, e.g. blast holes. The max number of composites is relatively large or unrestricted, to lower conditional bias.

Gaussian Distribution and Scaling

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

For all normal distributions, the probability to be within a certain fraction of the standard deviation is the same, no matter what the mean (location) or standard deviation (scale) is.

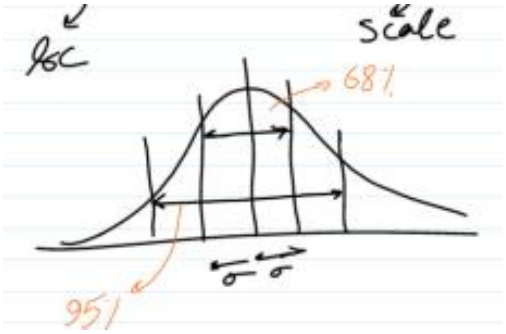


Figure 1. 68% within ±σ and 95% within ±2σ of the mean.

Given the symmetric nature of the Gaussian distribution, the standard normal distribution can be used to scale the probability to be within a deviation of the mean.

Example

Given: the probability to be within ±V of the mean → P. **Asked:** the probability if the mean remains the same but the variance is scaled by num_faces.

Step 1) With the assumption of Gaussianity, the probability can be mapped to a deviation from the mean for an N(0, 1) distribution and then scaled:

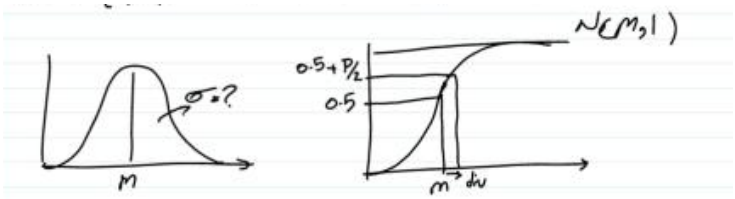


Figure 2. Mapping P to a deviation on the N(m, 1) CDF (0.5 + P/2).

dev = norm.ppf((1 + P) / 2)

std = 1 ↔ dev; std = ? ↔ V ⇒ std = V / dev, variance = $(V / dev)^2$

Step 2) Scale the variance based on the number of faces:

variance_new = variance / num_faces ⇒ σ_new = V / (dev·√num_faces)

Step 3) Get the probability to be within the deviation.

Note: since the mean remains the same, absolute and relative deviation are treated the same. With std = V / (dev·√num_faces), the deviation V maps to dev·√num_faces on N(0, 1).

Step 4) Now that the deviation from the mean is scaled to N(0, 1), get the CDF and calculate the probability:

$$(\text{norm.cdf}(\text{dev}\cdot\sqrt{\text{num_faces}}) - 0.5) \times 2 \Rightarrow \text{scaled probability} = 2\cdot\text{norm.cdf}(\text{dev}\cdot\sqrt{\text{num_faces}}) - 1$$

- In order to scale the probability, there is no need to have the deviation.
- Since the scaling is averaging, with the mean remaining the same, absolute deviation and relative deviation (i.e. percentage) are treated the same and cancel out.
- If it is required to provide the scaled variance or std, then the deviation associated with the probability is required.
- If the aggregation is summation, the same scaling workflow works for relative uncertainty, because mean → num_faces × mean and std → std × √num_faces (summation and independent cases). However, for absolute uncertainty with summation, the absolute uncertainty grows as the support volume grows and the source volumes/variables are independent.

Using DGM

The alternative approach is to get one base panel distribution and then scale the variance using the DGM. For example, monthly → quarterly with f = 1/3:

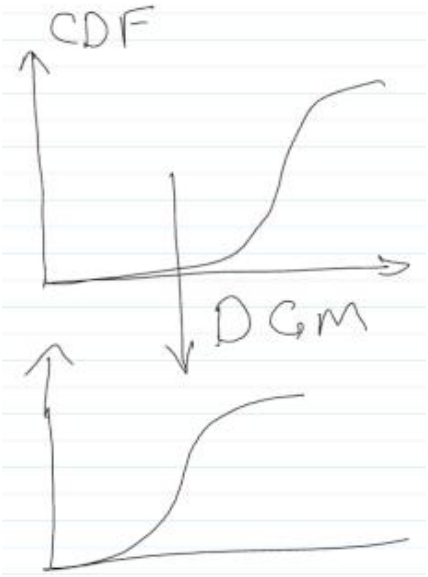


Figure 3. Monthly CDF scaled to quarterly with the DGM.

Kriging Plan

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Kriging models are often used for long, mid and short term planning. Depending on the purpose/application, kriging plans/scenarios are assessed. One of the main criteria is to match production data/models.

Some of the main parameters:

- Multiple passes → avoid an over-smooth model while mitigating conditional bias
- Search range
- Max and min composites
- Max composites per drill hole

A larger search and more min and max composites result in a SMOOTHER model and lower conditional bias.

Smooth	Restricted / variable
Lower variance / σ	Higher variance / σ
Lower conditional bias	Higher conditional bias
Underestimates grade in reconciliation over large panels	Overestimates grade (likewise)
Results in higher ore tonnage → more blocks above cut-off	Lower ore tonnage

Additional comment about the mineralization:

Structurally controlled	Disseminated
More structure and continuity	Higher variability and potentially larger nugget
More local accuracy	Less local uncertainty
Less volume variance and relatively higher uncertainty in larger volumetric supports	Lower uncertainty in larger volume supports (i.e. panels)
More value gained from additional drill holes	

Variogram

The effect of the variogram is tied to the value of information and local accuracy. In general, a more continuous variogram (particularly at short scales) would benefit from more data and therefore a larger search/more composites.

Data spacing

Data spacing in general is tied to uncertainty. With denser data (lower data spacing), there is less chance of over-smoothing, and therefore a larger search/more composites are deemed beneficial for reconciliation.

In fact, a multi-pass search is a way to control the kriging plan at different data spacings.

Metallurgy

OneNote notebook RMS / section Mining. Transcribed from handwritten notes and a pasted reference.

Comminution variables/indices

- **BWI:** Bond work index
- **DWI:** Drop weight index

RQD (Rock Quality Designation)

Summary of the pasted reference: RQD measures the quality of rock core from a borehole, expressed as a percentage that reflects the degree of jointing or fracturing in the rock mass. High RQD indicates good quality hard rock; low RQD indicates weathered, poor quality rock.

Rock quality	RQD (%)
Very poor (completely weathered rock)	< 25
Poor (weathered rocks)	25 to 50
Fair (moderately weathered rocks)	51 to 75
Good (hard rock)	76 to 90
Very good (fresh rocks)	91 to 100

Source linked on the page: theconstructor.org (RQD calculation article).

Mineral Scale

OneNote notebook RMS / section Mining. The page holds a single pasted reference figure.

The page shows the different scales at which the components of a deposit can be described, using a nickel laterite deposit as the example: elements → compounds → minerals → rocks → facies.

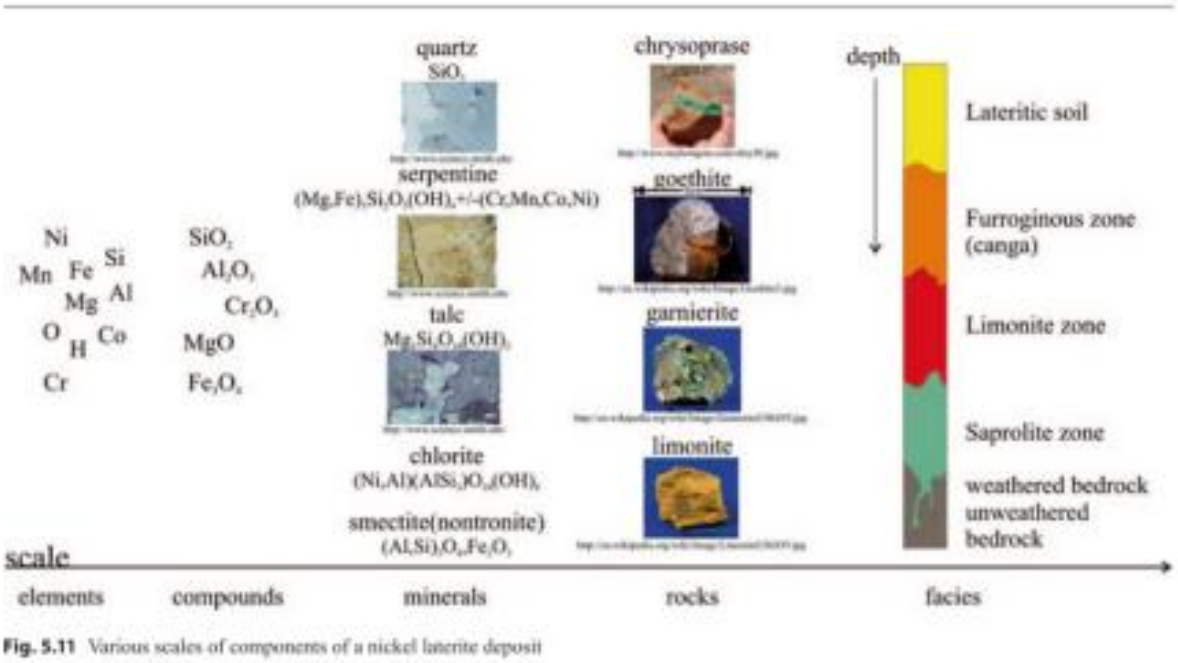


Fig. 5.11 Various scales of components of a nickel laterite deposit

Figure 1. Scales of components of a nickel laterite deposit (pasted reference figure): elements (Ni, Fe, Mg, Si, ...), compounds (SiO_2, MgO, Fe_2O_3, ...), minerals (quartz, serpentine, talc, chlorite, smectite), rocks (chrysoprase, goethite, garnierite, limonite) and facies with depth (lateritic soil, ferruginous zone, limonite zone, saprolite zone, weathered and unweathered bedrock).

Monthly, Quarterly and Annual upscaling

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

In order to assess the confidence and classify a mining reserve as inferred, indicated and measured, monthly, quarterly and often annual scale variability is assessed against data spacing.

It is often not feasible to upscale into quarterly and annual scale given the model size and the plant intake capacity. In such cases, the following assumptions are considered to infer the scale variance:

- The distribution of uncertainty for a monthly block is Gaussian, meaning that the distribution of values at a monthly block over the 100 realizations is Gaussian. This is a reasonable assumption at this scale based on the Central Limit Theorem.
- The material distributions of months within a quarter or annual scale are independent. This is a reasonable assumption based on the variogram ranges.
- The monthly blocks within each quarter or annual scale have approximately the same variability.

Different companies use different criteria, but it is well established to use quarterly and annual scale to classify resources as indicated or inferred. For example, within 15% of the estimate 90% of the time at annual scale can be considered as indicated. The corresponding data spacing is then used to classify the model.

Notes about variance

1) $\mathrm{Var}[x] = E[(x - \bar{x})^2] = E[x^2] - E[x]^2$

2) $\mathrm{Var}[ax] = E[a^2x^2] - a^2E[x]^2 = a^2\,\mathrm{Var}(x)$

3) $\mathrm{Var}(\Sigma\, a_i x_i) = \Sigma\, a_i^2\,\mathrm{Var}(x_i) + 2\,\Sigma_{i\neq j}\, a_i a_j\,\mathrm{Cov}(x_i, x_j) = \Sigma_i\,\Sigma_j\, a_i a_j\,\mathrm{Cov}(x_i, x_j)$ (linear combination)

independent case: $= \Sigma\, a_i^2\,\mathrm{Var}(x_i)$

4) Variance of upscaled values, e.g. upscale by averaging 4 random variables that are independent:

$\mathrm{Var}[\,¼x_1 + ¼x_2 + ¼x_3 + ¼x_4\,] = (1/16)(\mathrm{Var}(x_1) + \mathrm{Var}(x_2) + \mathrm{Var}(x_3) + \mathrm{Var}(x_4))$

if $\mathrm{Var}[x_i] = \sigma^2$ for i = 1..4: $\mathrm{Var}[\,¼\,\Sigma\, x_i\,] = (1/16)\cdot 4\sigma^2 = ¼\,\sigma^2$

Thus, for monthly to quarterly and annual averages, if the monthly blocks within each quarter or annual panel are independent:

$\mathrm{Var}_{\mathrm{quarterly}} = (1/3)\,\mathrm{Var}_{\mathrm{monthly}}$, $\mathrm{Var}_{\mathrm{annual}} = (1/12)\,\mathrm{Var}_{\mathrm{monthly}}$ (divisor = number of faces)

Thus, after numerical upscaling to monthly and inferring the local uncertainty at monthly scale, the monthly variance can be scaled down to quarterly and annual with the mean remaining the same. Then, the probability to be within 15% of the mean can be calculated.

Sum vs average to aggregate

If the aggregation to larger volumes is a "summation", then it should be considered that the variance and the mean both grow:

$$\mathrm{Var}[\,\Sigma_{i=1..3}\, x_i\,] = \Sigma\,\Sigma\, C_{ij} = 3\,\mathrm{Var}[x] \text{ (independent and same variance)};\ E\{\,\Sigma\, x_i\,\} = 3\,E[x]$$

Therefore, for sum: mean_upscale = num_faces × mean; std_upscale = √num_faces × std.

Compare this to "averaging" for aggregation with independent cases: mean_upscale = mean; std_upscale = std / √num_faces.

$$CV_{\text{upscale, sum}} = \sqrt{\text{num_faces}}\cdot\text{std} / (\text{num_faces}\cdot\text{mean}) = CV / \sqrt{\text{num_faces}}$$

$$CV_{\text{upscale, average}} = (\text{std} / \sqrt{\text{num_faces}}) / \text{mean} = CV / \sqrt{\text{num_faces}}$$

Therefore, analytical upscaling for both averaging and summation aggregations gives the same relative uncertainty.

Notes: with the variance operator, we know that $\mathrm{Var}[\Sigma\, x_i] = \Sigma\,\Sigma\, C_{ij}$. For fully dependent cases, smaller volumes have relatively higher variance due to spatial continuity and less variability. This is why, with both summation and averaging, larger support volumes result in lower relative uncertainty.

- **Sum:** the mean grows faster than the variance as the support volume increases.
- **Average:** the mean is constant but the variance is reduced by averaging, and as the support volume gets bigger that reduction is greater, as it covers more parts of the spatial variability.

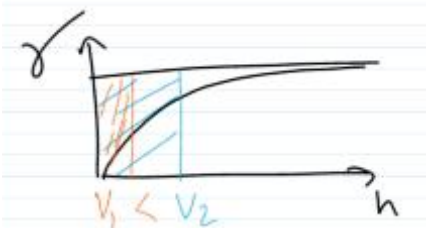


Figure 1. Variogram with support volumes $V_1 < V_2$.

Net Smelter Return

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Net smelter return (NSR) is the revenue that the owner of a mining property receives from the sale of the mine's metal/non-metal products, less transportation and refining costs.

Smelting: the process of heating ore to extract metals.

Open Pit Mining

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

This note contains a collection of notes and learnings about open pit mining.

- Open pit is a type of "surface" mining.
- The "bench height" is an important measure for open-pit mining.

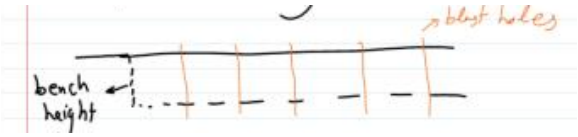


Figure 1. Bench height and blast holes.

The blasted material is excavated in one or multiple passes. For example, an SMU size of 15 m × 15 m × 7.5 m and a bench height of 15 m, to excavate in 2 passes.

The page also links an article on pushbacks in open pit mines ("What are pushbacks and why do we really need them in open pit mines?"):

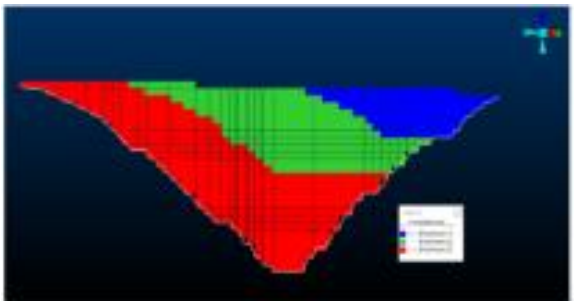


Figure 2. Pit cross-section showing pushback phases (from the linked article).

Regular spacing to 3 hole spacing

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Consider a regular spacing of a × b m.

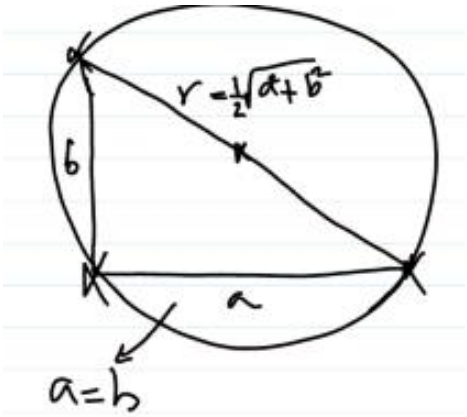


Figure 1. Right-angle triangle of three holes inscribed in a circle of radius r = ½√(a^2 + b^2).

For any right-angle triangle, the hypotenuse is the diameter of the circumscribed circle.

r (to include 3 holes) = ½ √(a^2 + b^2)

Resource Classification

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

Resource

A potentially valuable concentration of minerals (organic/inorganic) in or on the earth's crust.

Reserve (ore reserves)

Reserve refers to the economically mineable part of an indicated or measured mineral resource.

> Note that ore or waste refers to blocks of rock/material (SMU, selective mining unit) classified based on a grade cut-off. Ore is a body of rock with enough concentration of valuable minerals (e.g. gold) to be above the cut-off.

Inferred, Indicated and Measured

These terms refer to the level of confidence/uncertainty with which a block of rock (SMU) is identified as ore. This classification extends to identifying the reserve parts of a resource.

Definitions from the pasted training slide (summarized):

- **Mineral Resource:** a concentration of solid material of economic interest in or on the earth's crust, with reasonable prospects for eventual economic extraction.
- **Mineral Reserve:** the economically mineable part of a Measured and/or Indicated Mineral Resource; requires at least a PFS-level study (or operation) demonstrating positive cash flow.
- **Preliminary Economic Assessment (PEA):** a study of the economic viability of a mineral resource for a mining project; may use some or all of the Measured, Indicated and Inferred resources.
- **Pre-Feasibility Study (PFS):** a comprehensive study where the mining and processing method is understood using reasonable assumptions; includes modifying factors applied to Measured and Indicated resources only.
- **PEA Mineable Resources:** the portion of the mineral resources used in the PEA mine plan; may include modifying factors.

Scale, combined grade and Variability

OneNote notebook RMS / section Mining. Transcribed from handwritten notes.

In mining, it is often required to assess variability or uncertainty (local/global) at different scales. It is intuitive that by increasing the block scale and averaging, the uncertainty is reduced.

It is important to understand that measures like standard deviation and variance are measures of absolute variability/uncertainty. For example:

Var[Cu] = 1.5, Var[Zn] = 2

Var[10 Cu] = 10^2 Var[Cu], Var[2 Zn] = 2^2 Var[Zn]

If the Cu price is three times the Zn price, the combined equivalent grade is Cu + Zn/3, or 3Cu + Zn.

The variance of the combined grade will be different and can be found based on the variance of a linear combination:

$$Var = \Sigma_{i=1..n} \Sigma_{j=1..n} a_i a_j C_{ij}$$

However, a more reliable measure of uncertainty is a relative measure, such as CV = σ / m, or the probability to be within 15% of the mean. A measure like this accounts for the proportional effect of the mean on variability.